

Semi-Supervised Generative Classification via a Weighted Unlabeled Likelihood

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Abstract

We study semi-supervised binary classification in settings where labeled positives and negatives are scarce but unlabeled feature vectors are plentiful. We propose a two-component Gaussian mixture model estimated by a weighted likelihood that combines the fully observed class-conditional log-likelihoods with a tunable multiple λ of the unlabeled marginal mixture log-likelihood, fit by expectation-maximization with closed-form updates. The central contribution is an analysis of λ as an object of study in its own right. We prove an implicit-function-theorem characterization of the pseudo-true parameter path $\theta^*(\lambda)$: the derivative $d\theta^*/d\lambda$ equals the negative inverse Hessian times the unlabeled score residual, which is zero under correct specification and nonzero—and computable—under misspecification. This formula yields a computable pre-fitting score, the *alignment coefficient* $\mathcal{A}(0)$, whose sign predicts whether adding unlabeled data will improve or degrade the estimator. We characterize $\mathcal{A}(0)$ as a *decision support tool* rather than a universal oracle: in the small-labeled-data regime (the setting where semi-supervised learning is most consequential), the binary use/discard decision based on the sign of $\hat{\mathcal{A}}(0)$ achieves 65–72% accuracy with mean regret below 0.01 AUROC; as labeled data grow, regret collapses below 0.001 regardless of the decision, making the diagnostic most useful exactly when the decision matters most. We further propose a confidence-weighted generalization $\lambda(x)$ that downweights ambiguous unlabeled points, prove a component-wise score-attenuation property under a directional coherence condition, and implement it via an MM surrogate with closed-form updates at each iteration. Theorem 1 additionally yields a gradient-based procedure for *learning* λ from a validation set, reducing λ -selection to gradient ascent rather than grid search. Simulations across four estimation regimes—unsupervised mixture, small-sample supervised, global- λ semi-supervised, and confidence-weighted semi-supervised—demonstrate progressive improvement in parameter recovery and classification accuracy. We further validate on five UCI benchmarks, showing that the score-residual norm $\|\hat{g}_0\|$ correctly identifies datasets where the Gaussian assumption fails and semi-supervised learning is inadvisable. An open-source Python implementation is available as the `semi_supervised_em` package.

1 Introduction

Many applied classification problems share the same structural constraint: labels are expensive but features are abundant. A clinician can pull millions of electronic records but has time to adjudicate only hundreds. An insurer can process millions of claim events but has expert labels for a small fraction of matches. A sports analyst can observe thousands of player-possession observations but can verify only a handful by hand. In each case, the practitioner faces a partially labeled problem: a small labeled set that anchors the decision boundary and a large unlabeled corpus whose structure may sharpen or distort it.

Purely supervised models discard the unlabeled structure entirely and face high-variance estimation when labels are few. Fully unsupervised mixture fitting ignores the labeled anchors and can converge to components that are uninformative for the classification task. Semi-supervised

approaches attempt to exploit both, but the literature offers a wide spectrum of methods—from generative likelihood-based models at one end to discriminative consistency regularization at the other—with different assumptions about how unlabeled data relate to the task.

This paper proposes and analyzes a simple, transparent semi-supervised generative classifier. We model the class-conditional distributions as multivariate Gaussians and estimate parameters by maximizing a *weighted* log-likelihood,

$$\mathcal{J}_\lambda(\theta) = \underbrace{\ell_{\text{sup}}(\theta)}_{\text{supervised}} + \lambda \underbrace{\ell_{\text{unl}}(\theta)}_{\text{unlabeled mixture}}, \quad (1)$$

where $\lambda > 0$ is a scalar weight that controls how much the unlabeled corpus influences the fit. This objective is maximized by expectation-maximization (EM), which yields closed-form parameter updates in each iteration.

The weight λ is not merely a tuning nuisance; it is the conceptual spine of the paper. Increasing λ from zero continuously deforms the estimator from the purely supervised maximum likelihood estimator (MLE) toward an unlabeled-mixture-dominated fit. Each unit of λ can be interpreted as contributing an *effective* unlabeled sample budget of λN_u additional soft observations, partitioned across classes according to the current responsibilities. This effective-sample-size interpretation makes λ actionable: a practitioner can set it by asking how many additional pseudo-labeled examples the unlabeled corpus should count for, rather than treating it as an opaque regularization coefficient.

Under model misspecification—when the true class-conditional distributions are not Gaussian—the estimator converges to a λ -dependent pseudo-true parameter that can differ from the supervised pseudo-true parameter and may move the decision boundary in an unhelpful direction. We characterize this dependence and show that it motivates treating λ as a diagnostic as well as a tuning parameter: if increasing λ degrades labeled-set performance, the unlabeled structure is inconsistent with the Gaussian model and caution is warranted.

Motivating application. Our primary application is note-to-claimant matching at an insurance company. Each claim generates a set of notes (free text or structured records), each of which must be matched to one of several candidate claimants within the claim. Expert labels are available for a small fraction of note-claimant pairs, while a much larger corpus of feature-described pairs has no label. At inference time, the model produces a score $s(x) = \Pr(z = 1 \mid x; \hat{\theta})$ for each candidate pair, and scores are renormalized within each claim to enforce competition among candidates. This within-claim renormalization converts global posteriors into a proper ranking over candidates.

Contributions.

1. A weighted semi-supervised Gaussian mixture objective with unlabeled weight λ as the interpretable control parameter, fit by EM with closed-form updates (Section 3).
2. A formal effective-sample-size interpretation of λ : the fixed-point estimator is a weighted average of labeled sufficient statistics and responsibility-weighted unlabeled sufficient statistics (Proposition 2).
3. An implicit-function-theorem theorem (Theorem 1) giving the exact derivative $d\theta^*(\lambda)/d\lambda$ in terms of the unlabeled score residual at the supervised pseudo-true parameter—zero under correct specification, nonzero and computable under misspecification—plus a sample diagnostic $\hat{g}_0$ that can be evaluated before fitting.

4. A confidence-weighted generalization $\lambda(x_j) = \lambda \cdot (\max_g \gamma_j^{(g)})^\alpha$ implemented via an MM surrogate with closed-form updates, and a proven component-wise score-attenuation property under a directional coherence condition (Proposition 4).
5. A gradient-based procedure for *learning* λ from data (Section 3.8): the IFT derivative (10) yields a closed-form alignment coefficient $\mathcal{A}(0)$ whose sign provides a decision-support diagnostic for whether to use unlabeled data. In the small-label regime, the binary decision $D = \mathbf{1}[\hat{\mathcal{A}}(0) > 0]$ achieves 65–72% accuracy with mean regret below 0.01 AUROC; the diagnostic is most reliable precisely when the unlabeled contribution is largest and the decision is most consequential.
6. Simulation evidence across four regimes (unsupervised, supervised, global- λ , confidence-weighted) on parameter recovery, classification accuracy, calibration, and robustness to misspecification.
7. Empirical validation on five UCI binary classification benchmarks, showing that the score-residual norm $\|\hat{g}_0\|$ correctly identifies the near-Gaussian datasets (Banknote, Breast Cancer) where the diagnostic achieves 65–70% accuracy from the non-Gaussian ones (Ionosphere, Sonar, Spambase) where it falls to 40–50%.
8. An open-source Python package, `semi-supervised-em`, with a scikit-learn-compatible API.

The unifying thread. The score residual $\hat{g}_0$ —the gradient of the unlabeled log-likelihood evaluated at the supervised MLE—is the paper’s load-bearing object, but its two roles must be carefully distinguished.

Its norm $\|\hat{g}_0\|$ is a pre-fitting indicator of *magnitude of mismatch*: a large norm means the unlabeled corpus is structurally inconsistent with the Gaussian model and that the parameter path will move rapidly as λ grows. It does not by itself say whether that movement helps or hurts.

Its direction, via the alignment coefficient $\mathcal{A}(0) = -\nabla_{\theta} \ell_V(\hat{\theta}(0))^\top \hat{H}(0)^{-1} \hat{g}_0$, is the *directional* diagnostic: if $\mathcal{A}(0) > 0$, the unlabeled data pull the estimator toward the validation optimum; if $\mathcal{A}(0) < 0$, they pull it away. Gradient ascent on λ increases validation likelihood if and only if $\mathcal{A}(\lambda) > 0$.

The single formula $d\theta^*/d\lambda = -H^{-1}g_0$ thus serves simultaneously as magnitude certificate (via $\|H^{-1}g_0\|$), directional diagnostic (via its inner product with the validation gradient), and optimization oracle—computed in one Hessian solve at essentially no additional cost beyond the supervised fit. Corollary 1.3 further shows that $\mathcal{A}(0)$ predicts the first-order change in validation likelihood, with the second-order term explaining why the prediction degrades at larger λ .

Paper outline. Section 2 reviews related work. Section 3 defines the model and EM algorithm. Section 4 develops the theoretical analysis. Sections 5 and 6 present simulation and empirical results. Section ?? discusses scope and limitations.

2 Related Work

2.1 Semi-Supervised Learning

Semi-supervised learning has a long history; comprehensive surveys include ? and ?. Methods divide broadly into generative and discriminative families. Generative semi-supervised methods learn a joint model of features and labels and use unlabeled examples to improve the marginal feature distribution estimate. The classical example is EM for Gaussian mixtures with partially observed class labels (??). Discriminative methods—including consistency regularization (??),

pseudo-labeling (?), and self-training (?)—add unlabeled data through surrogate losses or teacher-student loops without committing to a generative model.

Our method is generative and likelihood-based, placing it closest to the tradition of ? and ?. ? use a related downweighting of unlabeled data in naive Bayes text classification; they note empirically that reducing the unlabeled weight can improve performance when the generative assumptions are violated, but do not derive the effective-sample-size interpretation or the pseudo-true parameter path.

2.2 Gaussian Mixture Models, Weighting, and Misspecification

Two-component Gaussian mixture models are a canonical probabilistic classification tool (??). The maximum likelihood problem for partially labeled Gaussian mixtures has been studied by ? and ?, and ? develop model-based clustering via EM for the fully unlabeled case.

Closest method-level comparator. ? study semi-supervised Gaussian mixture models for phonetic classification with an explicit relative weighting between labeled and unlabeled log-likelihoods, including empirical analysis of critical weight values and how they vary with labeled sample size. Our paper is complementary: we develop a focused analytical treatment of the same weighting parameter in the simplest Gaussian generative binary classifier, contributing (i) the implicit-function-theorem characterization of the pseudo-true parameter path, (ii) the closed-form score-residual diagnostic $\hat{g}_0$, and (iii) the confidence-weighted extension with its provable damage bound—none of which appear in ?.

Closest conceptual comparator. ? study semi-supervised hidden Markov models via a homotopy approach in which a parameter continuously interpolates between supervised ($\lambda = 0$) and unsupervised ($\lambda = 1$) objectives, and discuss performance degradation under misspecification. Our Theorem 1 provides an estimator-level companion to that interpolation idea: rather than tracking the optimization path numerically, we give a formula for how the fixed-point target moves as λ varies.

Misspecification. ? show rigorously that unlabeled data can *decrease* the accuracy of generative classifiers under model misspecification. Their result is qualitative; our Theorem 1 provides a quantitative companion, giving the rate and direction of estimand damage as a function of λ via the score residual g_0 . The confidence-weighted $\lambda(x)$ in Section 3.4 is a direct algorithmic response: under a directional coherence condition it attenuates each coordinate of the score residual, reducing per-coordinate misspecification exposure (Proposition 4).

2.3 Positive-Unlabeled and Weakly Supervised Learning

Positive-unlabeled (PU) learning (??) considers a setting where only positive labels are observed and the negative class must be inferred from the unlabeled pool. Our setting is distinct: we observe labeled positives *and* labeled negatives, so the supervised component anchors both class-conditional distributions. When labeled negatives are sparse, our method naturally degrades gracefully because the unlabeled likelihood can inform the negative component; this is a point of contact with PU learning, but the formal setup differs.

Weakly supervised methods (??) use programmatic or noisy labels rather than unlabeled examples per se. Our unlabeled examples carry no label information at all; they enter only through the generative model’s marginal likelihood.

2.4 Entity Matching and Candidate Ranking

The notes-to-claimants matching problem belongs to the entity matching and record linkage literature (??). Standard approaches include rule-based blocking, string-similarity classifiers, and learning-to-rank methods. Semi-supervised generative models have been applied to record linkage by ?, who use EM on the Fellegi-Sunter model. Our method differs in operating on arbitrary numeric feature vectors rather than agreement patterns, and in explicitly modeling within-claim competition through post-hoc score renormalization.

3 Method

3.1 Problem Setting and Notation

We observe feature vectors $x \in \mathbb{R}^m$. Data are partitioned into labeled positives ($z = 1$), labeled negatives ($z = 0$), and unlabeled examples:

$$\mathcal{D}_1 = \{x_i^{(1)}\}_{i=1}^{N_1}, \quad \mathcal{D}_0 = \{x_i^{(0)}\}_{i=1}^{N_0}, \quad \mathcal{U} = \{x_j^{(u)}\}_{j=1}^{N_u}.$$

Let $N_{\text{obs}} = N_0 + N_1$. We assume all observations are drawn from a two-component Gaussian mixture,

$$x \sim \pi \mathcal{N}(\mu_1, \Sigma_1) + (1 - \pi) \mathcal{N}(\mu_0, \Sigma_0),$$

and that the labeled sets are draws conditional on their respective class labels. We write parameters collectively as $\theta = (\pi, \mu_1, \mu_0, \Sigma_1, \Sigma_0)$, where for $g \in \{0, 1\}$:

$$x \mid z = g \sim \mathcal{N}(\mu_g, \Sigma_g), \quad \Pr(z = 1) = \pi, \quad \Pr(z = 0) = 1 - \pi.$$

The multivariate Gaussian density is

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{m/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right).$$

3.2 Weighted Semi-Supervised Objective

We maximize a *weighted* log-likelihood combining labeled and unlabeled examples with unlabeled weight $\lambda > 0$:

$$\begin{aligned} \mathcal{J}_\lambda(\theta) = & \sum_{i=1}^{N_1} \log\left[\pi \mathcal{N}(x_i^{(1)} \mid \mu_1, \Sigma_1)\right] + \sum_{i=1}^{N_0} \log\left[(1 - \pi) \mathcal{N}(x_i^{(0)} \mid \mu_0, \Sigma_0)\right] \\ & + \lambda \sum_{j=1}^{N_u} \log\left(\pi \mathcal{N}(x_j^{(u)} \mid \mu_1, \Sigma_1) + (1 - \pi) \mathcal{N}(x_j^{(u)} \mid \mu_0, \Sigma_0)\right). \end{aligned} \quad (2)$$

The labeled terms use the *joint* class likelihood $p_\theta(x, z)$, not the class-conditional $p_\theta(x \mid z)$, so π is identified by the labeled class counts as well as by the unlabeled responsibilities. When $\lambda \rightarrow 0$ we recover the purely supervised joint MLE; when λ is large, the unlabeled mixture log-likelihood dominates.

3.3 EM Algorithm

Direct maximization of $\mathcal{J}_\lambda$ is intractable because of the log-sum in the unlabeled term. We instead apply the EM algorithm, treating the latent class labels $z_j^{(u)} \in \{0, 1\}$ of unlabeled examples as missing data.

Define responsibilities

$$\gamma_j := \Pr(z_j^{(u)} = 1 \mid x_j^{(u)}; \theta) = \frac{\pi \mathcal{N}(x_j^{(u)} \mid \mu_1, \Sigma_1)}{\pi \mathcal{N}(x_j^{(u)} \mid \mu_1, \Sigma_1) + (1 - \pi) \mathcal{N}(x_j^{(u)} \mid \mu_0, \Sigma_0)}.$$

Define the *soft counts*

$$\tilde{N}_1 = N_1 + \lambda \sum_{j=1}^{N_u} \gamma_j, \quad \tilde{N}_0 = N_0 + \lambda \sum_{j=1}^{N_u} (1 - \gamma_j), \quad \tilde{N} = N_{\text{obs}} + \lambda N_u.$$

E-step. Given $\theta^{(t)}$, update responsibilities γ_j for all unlabeled points. Compute in log-space for numerical stability:

$$\log r_{1j} = \log \pi + \log \mathcal{N}(x_j^{(u)} \mid \mu_1, \Sigma_1), \quad \log r_{0j} = \log(1 - \pi) + \log \mathcal{N}(x_j^{(u)} \mid \mu_0, \Sigma_0),$$

$$\gamma_j = \exp(\log r_{1j} - \text{LSE}(\log r_{1j}, \log r_{0j})),$$

where $\text{LSE}(a, b) = \max(a, b) + \log(e^{a-\max} + e^{b-\max})$.

M-step. Maximize the expected weighted complete-data log-likelihood to obtain closed-form updates:

$$\pi^{\text{new}} = \frac{\tilde{N}_1}{\tilde{N}}, \tag{3}$$

$$\mu_1^{\text{new}} = \frac{\sum_{x \in \mathcal{D}_1} x + \lambda \sum_{j=1}^{N_u} \gamma_j x_j^{(u)}}{\tilde{N}_1}, \quad \mu_0^{\text{new}} = \frac{\sum_{x \in \mathcal{D}_0} x + \lambda \sum_{j=1}^{N_u} (1 - \gamma_j) x_j^{(u)}}{\tilde{N}_0}, \tag{4}$$

$$\Sigma_1^{\text{new}} = \frac{\sum_{x \in \mathcal{D}_1} (x - \mu_1^{\text{new}})(x - \mu_1^{\text{new}})^\top + \lambda \sum_{j=1}^{N_u} \gamma_j (x_j^{(u)} - \mu_1^{\text{new}})(x_j^{(u)} - \mu_1^{\text{new}})^\top}{\tilde{N}_1} + \epsilon \mathbf{I}, \tag{5}$$

$$\Sigma_0^{\text{new}} = \frac{\sum_{x \in \mathcal{D}_0} (x - \mu_0^{\text{new}})(x - \mu_0^{\text{new}})^\top + \lambda \sum_{j=1}^{N_u} (1 - \gamma_j) (x_j^{(u)} - \mu_0^{\text{new}})(x_j^{(u)} - \mu_0^{\text{new}})^\top}{\tilde{N}_0} + \epsilon \mathbf{I}, \tag{6}$$

where $\epsilon > 0$ is a small ridge constant that enforces positive definiteness.

Derivation sketch. Form the expected weighted complete-data log-likelihood $\mathcal{Q}_\lambda(\theta; \theta^{(t)}) = \mathbb{E}[\log p(X, Z; \theta) \mid X; \theta^{(t)}]$ where unlabeled contributions are weighted by λ , take partial derivatives with respect to each parameter, set to zero, and solve. Equations (3)–(6) are the unique maximizers under standard regularity (positive-definite covariances).

Algorithm 1 Weighted Semi-Supervised EM

Require: $\mathcal{D}_1, \mathcal{D}_0, \mathcal{U}$, weight $\lambda > 0$, tolerance ε , ridge ϵ

- 1: Initialize $\mu_g^{(0)}, \Sigma_g^{(0)}, \pi^{(0)}$ from supervised MLEs; set $\gamma_j^{(0)} = \pi^{(0)}$
 - 2: **repeat**
 - 3: **E-step:** Compute γ_j via log-sum-exp for all j
 - 4: **M-step:** Update π, μ_g, Σ_g via (3)–(6)
 - 5: **until** max parameter change $< \varepsilon$ **return** $\hat{\theta} = (\hat{\pi}, \hat{\mu}_1, \hat{\mu}_0, \hat{\Sigma}_1, \hat{\Sigma}_0)$
-

Convergence and initialization. Repeat E/M steps until

$$\max \left\{ \|\mu_1^{\text{new}} - \mu_1\|_\infty, \|\mu_0^{\text{new}} - \mu_0\|_\infty, \|\Sigma_1^{\text{new}} - \Sigma_1\|_\infty, \|\Sigma_0^{\text{new}} - \Sigma_0\|_\infty, |\pi^{\text{new}} - \pi| \right\} < \varepsilon.$$

A robust warm start uses the supervised MLEs:

$$\mu_g^{(0)} = \frac{1}{N_g} \sum_{x \in \mathcal{D}_g} x, \quad \Sigma_g^{(0)} = \frac{1}{N_g} \sum_{x \in \mathcal{D}_g} (x - \mu_g^{(0)})(x - \mu_g^{(0)})^\top + \epsilon \mathbf{I}, \quad \pi^{(0)} = \frac{N_1}{N_{\text{obs}}}.$$

Unlabeled responsibilities are initialized as $\gamma_j^{(0)} = \pi^{(0)}$. This ensures that EM begins from a configuration that is already consistent with the labeled data, reducing sensitivity to initialization and accelerating convergence.

3.4 Confidence-Weighted Unlabeled Likelihood

The scalar λ treats all unlabeled points equally. We propose a point-wise generalization:

$$\mathcal{J}_{\lambda, \alpha}(\theta) = \ell_{\text{sup}}(\theta) + \lambda \sum_{j \in \mathcal{U}} w_j^{(\alpha)}(\theta) \log p_\theta(x_j^{(u)}), \quad (7)$$

where the confidence weight of unlabeled point j is

$$w_j^{(\alpha)}(\theta) = \left(\max_{g \in \{0,1\}} \gamma_j^{(g)}(\theta) \right)^\alpha, \quad \alpha \geq 0, \quad (8)$$

with $\gamma_j^{(1)} = \gamma_j$ and $\gamma_j^{(0)} = 1 - \gamma_j$. At $\alpha = 0$ we recover the uniform- λ objective. As $\alpha \rightarrow \infty$ only the most confidently classified unlabeled points contribute. For a point classified with certainty, $w_j^{(\alpha)} = 1$; for a maximally ambiguous point ($\gamma_j = 0.5$), $w_j^{(\alpha)} = 2^{-\alpha}$.

Algorithm under confidence weighting (MM surrogate). Because $w_j^{(\alpha)}(\theta)$ depends on θ , the objective (7) is *not* a standard weighted log-likelihood with fixed weights, and direct EM does not apply. We instead treat this as a majorization-minimization (MM) procedure: at iteration t , freeze the confidence weights at the current parameter,

$$\tilde{w}_j^{(t)} = \lambda w_j^{(\alpha)}(\theta^{(t)}),$$

and maximize the resulting surrogate objective—a weighted mixture log-likelihood with *fixed* point-weights $\tilde{w}_j^{(t)}$. The E-step computes responsibilities γ_j at $\theta^{(t)}$ as before. The M-step soft counts become

$$\tilde{N}_1^{(\alpha)} = N_1 + \sum_j \tilde{w}_j^{(t)} \gamma_j, \quad \tilde{N}_0^{(\alpha)} = N_0 + \sum_j \tilde{w}_j^{(t)} (1 - \gamma_j), \quad \tilde{N}^{(\alpha)} = N_{\text{obs}} + \sum_j \tilde{w}_j^{(t)},$$

and update equations (3)–(6) carry through with λ replaced by $\tilde{w}_j^{(t)}$ for point j . All M-step updates are closed-form for the surrogate. Proposition 1 (monotonicity) applies to each surrogate step but *not* globally across iterations, because the surrogate changes at each step when the weights are refreshed. In practice the algorithm converges reliably; we do not claim that it increases the true objective (7) at every step.

3.5 Numerical Considerations

- **Log-space evaluation.** Always compute Gaussian log-densities using $\log |\Sigma|$ (via slogdet) and quadratic forms via linear solves (e.g., Cholesky) to avoid under/overflow.
- **PSD enforcement.** The ridge $\epsilon \mathbf{I}$ with $\epsilon \approx 10^{-6}$ is applied each M-step; it is also part of the initialization.
- **Responsibility clipping.** Clip $\gamma_j \in [\delta, 1 - \delta]$ with $\delta \approx 10^{-6}$ to prevent degeneracy when one component wins too early.
- **Feature scaling.** Standardize features to unit scale before fitting to improve covariance conditioning.

3.6 Prediction and Within-Group Competition

For any new feature vector x , the posterior probability of the positive class is

$$\Pr(z = 1 \mid x; \hat{\theta}) = \frac{\hat{\pi} \mathcal{N}(x \mid \hat{\mu}_1, \hat{\Sigma}_1)}{\hat{\pi} \mathcal{N}(x \mid \hat{\mu}_1, \hat{\Sigma}_1) + (1 - \hat{\pi}) \mathcal{N}(x \mid \hat{\mu}_0, \hat{\Sigma}_0)}.$$

In matching problems where exactly one candidate per group should be selected, scores within group c can be renormalized:

$$\tilde{s}_c(x) = \frac{\Pr(z = 1 \mid x; \hat{\theta})}{\sum_{x' \in \text{group}_c} \Pr(z = 1 \mid x'; \hat{\theta})},$$

converting global posterior probabilities into within-group rank-preserving shares that sum to one. The renormalization does not change rankings within groups but ensures that each group’s scores constitute a valid distribution over candidates, which can be useful when downstream decisions require exactly one selection per group.

3.7 Choosing the Unlabeled Weight λ

Interpretation via effective sample size. The M-step updates show that unlabeled examples contribute to the sufficient statistics as if they provided $\lambda \sum_j \gamma_j$ soft additional samples to class 1 and $\lambda \sum_j (1 - \gamma_j)$ to class 0. Thus λN_u is an *approximate* effective unlabeled sample budget: the exact soft counts depend on the responsibilities $\sum_j \gamma_j$, which themselves depend on the converged $\theta^*(\lambda)$, so the ESS interpretation is exact only when classes are well-separated ($\sum_j \gamma_j \approx \pi N_u$). A simple heuristic default is

$$\lambda \approx c \cdot \frac{N_{\text{obs}}}{N_u}, \quad c \in [0.5, 2],$$

so unlabeled data carry total weight comparable to the labeled set, though empirical λ -selection (below) is preferred in practice.

Selection by labeled validation (preferred in practice). Hold out a labeled validation set $\mathcal{V}$ and fit over a logarithmic grid $\lambda \in \Lambda$, picking

$$\lambda^* \in \arg \max_{\lambda \in \Lambda} \text{Metric}(\Pr(z = 1 \mid x; \hat{\theta}_\lambda), z) \quad \text{on } \mathcal{V}.$$

Calibration-driven choice. If well-calibrated probabilities are required, choose λ that minimizes Expected Calibration Error (ECE) or Brier score on $\mathcal{V}$.

Information criteria. When labels are very scarce, one may approximate the penalized marginal likelihood of unlabeled data under $\hat{\theta}_\lambda$ and select λ by an ICL-type criterion, though this is less reliable than labeled validation.

3.8 Learning the Unlabeled Weight λ

The procedures above treat λ as a hyperparameter chosen externally. Theorem 1 enables something more principled: a closed-form expression for how any criterion changes as λ varies, giving gradient-based λ -selection without a grid search.

Setup. Partition labeled data into training $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_0$ and validation $\mathcal{V}$. Let $\hat{\theta}(\lambda)$ be the EM fixed point trained on $\mathcal{D} \cup \mathcal{U}$ at weight λ , and let $\ell_{\mathcal{V}}(\theta) = |\mathcal{V}|^{-1} \sum_{i \in \mathcal{V}} \log p_\theta(x_i, y_i)$ denote the mean labeled log-likelihood on the validation set. (Using a per-observation average makes the magnitude of $\mathcal{A}(0)$ independent of validation set size, easing threshold selection.) We seek $\lambda^* = \arg \max_{\lambda \geq 0} \ell_{\mathcal{V}}(\hat{\theta}(\lambda))$.

Gradient via IFT. The sample analog of Theorem 1 gives $\hat{\theta}'(\lambda) = -\hat{H}(\lambda)^{-1} \hat{g}(\hat{\theta}(\lambda))$, where $\hat{H}(\lambda)$ is the sample Hessian of $\mathcal{J}_\lambda$ at $\hat{\theta}(\lambda)$ and $\hat{g}(\theta) = N_u^{-1} \sum_{j \in \mathcal{U}} \nabla_\theta \log p_\theta(x_j^{(u)})$ is the sample unlabeled score. By the chain rule,

$$\frac{d}{d\lambda} \ell_{\mathcal{V}}(\hat{\theta}(\lambda)) = \nabla_\theta \ell_{\mathcal{V}}(\hat{\theta}(\lambda))^\top \hat{\theta}'(\lambda) = -\nabla_\theta \ell_{\mathcal{V}}(\hat{\theta}(\lambda))^\top \hat{H}(\lambda)^{-1} \hat{g}(\hat{\theta}(\lambda)). \quad (9)$$

The right-hand side is the inner product, in the curvature metric $\hat{H}(\lambda)^{-1}$, between the *unlabeled-induced parameter shift* $\hat{\theta}'(\lambda) = -\hat{H}(\lambda)^{-1} \hat{g}$ and the *validation score* $\nabla_\theta \ell_{\mathcal{V}}(\hat{\theta}(\lambda))$. Increasing λ improves validation performance if and only if these two vectors point in the same direction (in the $\hat{H}^{-1}$ metric).

Alignment coefficient. Define

$$\mathcal{A}(\lambda) := \underbrace{-\hat{H}(\lambda)^{-1} \hat{g}(\hat{\theta}(\lambda))}_{\text{unlabeled-induced shift } \hat{\theta}'(\lambda)} \cdot \underbrace{\nabla_\theta \ell_{\mathcal{V}}(\hat{\theta}(\lambda))}_{\text{validation score}}.$$

Then $d\ell_{\mathcal{V}}/d\lambda = \mathcal{A}(\lambda)$, and its sign is the decision:

- $\mathcal{A}(\lambda) > 0$: validation likelihood improves; parameter shift and validation score are *aligned*.
- $\mathcal{A}(\lambda) < 0$: validation likelihood deteriorates; they are *opposed*.
- $\mathcal{A}(\lambda) = 0$: first-order local optimality condition for λ .

Practical algorithm. Evaluate $\mathcal{A}(0)$ at the supervised MLE before running any semi-supervised EM.

1. Fit $\hat{\theta}(0)$ (supervised MLE on $\mathcal{D}$).
2. Compute $\hat{g}_0$, $\hat{H}(0)$, and $\mathcal{A}(0) = -\nabla_{\theta} \ell_{\mathcal{V}}(\hat{\theta}(0))^{\top} \hat{H}(0)^{-1} \hat{g}_0$.
3. If $\mathcal{A}(0) \leq 0$: set $\lambda = 0$. This is a local first-order condition at $\lambda = 0$; it does not rule out beneficial semi-supervised fitting when the criterion is non-monotone, but is the recommended starting decision.
4. If $\mathcal{A}(0) > 0$: initialize $\lambda^{(0)} > 0$ and iterate: run EM to convergence at $\lambda^{(k)}$, compute $\mathcal{A}(\lambda^{(k)})$, update $\lambda^{(k+1)} = \lambda^{(k)} + \eta \mathcal{A}(\lambda^{(k)})$, stop when $|\mathcal{A}(\lambda^{(k)})| < \varepsilon_{\lambda}$.

We use the population IFT derivative as a first-order approximation at the finite-sample fixed point; this is exact for the population problem and asymptotically valid for finite samples as $n \rightarrow \infty$ under standard regularity (consistency of $\hat{\theta}(\lambda) \rightarrow \theta^*(\lambda)$ and continuity of $H(\lambda)$; see Appendix ??).

Remark 1 (Population vs. sample objects). *Theorem 1 is a population statement: $g_0 = \nabla_{\theta} \bar{\ell}_{\text{unl}}(\theta^*(0))$ and $H(0) = \nabla_{\theta}^2 \bar{\mathcal{J}}_0(\theta^*(0))$ are population quantities. In practice we substitute sample analogs $\hat{g}_0 = N_u^{-1} \sum_j \nabla_{\theta} \log p_{\hat{\theta}(0)}(x_j^{(u)})$ and $\hat{H}(0)$, which are consistent by the law of large numbers. The resulting sample alignment coefficient $\hat{\mathcal{A}}(0)$ is a consistent estimator of the population gradient $d\ell_{\mathcal{V}}/d\lambda$, with estimation error $O_p(N_u^{-1/2} + N_1^{-1/2} + N_{\mathcal{V}}^{-1/2})$.*

Evaluating $\mathcal{A}(0)$ costs one supervised fit, one pass over $\mathcal{U}$ to form $\hat{g}_0$, and a single solve of a $\dim(\theta) \times \dim(\theta)$ linear system; the dominant cost is the supervised fit.

Remark 2 (Dual role of λ and connection to $\hat{g}_0$). *At $\lambda = 0$, the alignment coefficient collapses to $\mathcal{A}(0) = -\nabla_{\theta} \ell_{\mathcal{V}}(\hat{\theta}(0))^{\top} \hat{H}(0)^{-1} \hat{g}_0$. The diagnostic $\hat{g}_0$ of Remark 5 thus has two complementary uses: its norm $\|\hat{g}_0\|$ measures misspecification risk (Corollary 1.2), and its inner product with the validation gradient in the $\hat{H}(0)^{-1}$ metric determines whether increasing λ is beneficial. A large $\hat{g}_0$ opposed to the validation gradient signals that unlabeled data are harmful; a large $\hat{g}_0$ aligned with it signals they should be trusted. Together with Proposition 2, this gives λ a dual role: it controls the effective unlabeled sample size, and it is selected to align the unlabeled-induced parameter update with validation performance. The same object that quantifies how much unlabeled data count also determines whether they should count at all.*

4 Theory

This section develops four results. Proposition 1 establishes that the weighted EM algorithm monotonically increases the weighted observed-data log-likelihood. Proposition 2 formalizes the effective-sample-size interpretation of λ at any fixed point of the EM map. Theorem 1 is the main result: it gives the exact derivative of the pseudo-true parameter with respect to λ , showing how and at what rate model misspecification propagates into the estimand. Proposition 4 shows that the confidence-weighted generalization of Section 3.4 has a bounded-damage property.

4.1 EM Monotonicity

Proposition 1 (Weighted EM monotonicity). *Let $\theta^{(t)}$ be the parameter at iteration t and $\theta^{(t+1)}$ be the result of one E/M step applied to the base objective $\mathcal{J}_{\lambda}$ (2) with fixed λ . Then $\mathcal{J}_{\lambda}(\theta^{(t+1)}) \geq \mathcal{J}_{\lambda}(\theta^{(t)})$. This monotonicity holds for the base objective only; the confidence-weighted surrogate*

procedure of Section 3.4 does not satisfy global monotonicity because the surrogate changes at each iteration.

Proof. Define the weighted expected complete-data log-likelihood

$$\begin{aligned} \mathcal{Q}_\lambda(\theta; \theta^{(t)}) &= \sum_{i=1}^{N_1} \log[\pi \mathcal{N}(x_i^{(1)} \mid \mu_1, \Sigma_1)] + \sum_{i=1}^{N_0} \log[(1 - \pi) \mathcal{N}(x_i^{(0)} \mid \mu_0, \Sigma_0)] \\ &\quad + \lambda \sum_{j=1}^{N_u} \left[\gamma_j^{(t)} \log(\pi \mathcal{N}(x_j^{(u)} \mid \mu_1, \Sigma_1)) + (1 - \gamma_j^{(t)}) \log((1 - \pi) \mathcal{N}(x_j^{(u)} \mid \mu_0, \Sigma_0)) \right]. \end{aligned}$$

By the standard EM argument (?), for any θ and any fixed $\theta^{(t)}$:

$$\mathcal{J}_\lambda(\theta) - \mathcal{J}_\lambda(\theta^{(t)}) = \mathcal{Q}_\lambda(\theta; \theta^{(t)}) - \mathcal{Q}_\lambda(\theta^{(t)}; \theta^{(t)}) + \lambda \text{KL}(q^{(t)} \parallel p(\cdot \mid X_u; \theta))$$

where $q^{(t)}$ is the responsibility distribution at $\theta^{(t)}$ and the KL term is non-negative. The M-step chooses $\theta^{(t+1)} = \arg \max_\theta \mathcal{Q}_\lambda(\theta; \theta^{(t)})$, so $\mathcal{Q}_\lambda(\theta^{(t+1)}; \theta^{(t)}) \geq \mathcal{Q}_\lambda(\theta^{(t)}; \theta^{(t)})$, which together with the non-negative KL gap gives the result. $\square$

4.2 Effective Sample Size

Proposition 2 (Effective sample size at fixed point). *Let θ^* be a fixed point of the weighted EM map. Denote the fixed-point responsibilities $\gamma_j^* = \Pr(z_j^{(u)} = 1 \mid x_j^{(u)}; \theta^*)$ and the class-specific unlabeled soft counts $S_1^* = \sum_j \gamma_j^*$ and $S_0^* = \sum_j (1 - \gamma_j^*)$. Then the fixed-point estimators satisfy:*

$$\mu_g^* = \frac{N_g \bar{x}_g^{\text{sup}} + \lambda S_g^* \bar{x}_g^{\text{unl}}}{N_g + \lambda S_g^*},$$

where $\bar{x}_g^{\text{sup}}$ is the labeled sample mean for class g and $\bar{x}_g^{\text{unl}}$ is the responsibility-weighted unlabeled mean for class g . Consequently, the unlabeled data contribute λS_g^* effective additional labeled observations to class g 's mean estimator. The covariance fixed-point Σ_g^* has the same soft-count structure but involves deviations from the updated mean, so the effective-sample-size interpretation is analogous rather than exact. For π^* , the joint labeled terms contribute N_g counts directly (see (3)), so the ESS decomposition carries over to the mixing weight as well.

Proof. The fixed-point condition requires that θ^* satisfy the M-step equations with responsibilities computed at θ^* . Substituting the definition of $\tilde{N}_g$ into (4) and factoring gives the stated form directly. The contribution λS_g^* is the weight placed on the unlabeled sufficient statistic for class g . $\square$

Remark 3. *The effective contribution λS_g^* depends on the fixed-point responsibilities, making Proposition 2 a characterization of the fixed point rather than a simple linear function of λ . However, when the model is well-specified and the classes are well-separated, $S_1^* \approx \pi N_u$ and $S_0^* \approx (1 - \pi) N_u$, so the effective budget is approximately $\lambda \pi N_u$ for class 1 and $\lambda (1 - \pi) N_u$ for class 0. The default $\lambda \approx N_{\text{obs}}/N_u$ then balances labeled and unlabeled contributions.*

4.3 The Pseudo-True Parameter Path

Let P_0 denote the true data-generating distribution, which need not be a Gaussian mixture. Define the population weighted objective

$$\bar{\mathcal{J}}_\lambda(\theta) = \underbrace{\mathbb{E}_{P_0}[\ell_{\text{sup}}(\theta)]}_{\bar{\ell}_{\text{sup}}(\theta)} + \lambda \underbrace{\mathbb{E}_{P_0}[\ell_{\text{unl}}(\theta)]}_{\bar{\ell}_{\text{unl}}(\theta)},$$

where $\bar{\ell}_{\text{sup}}$ is the expectation over the labeled class-conditional distributions of P_0 and $\bar{\ell}_{\text{unl}}$ is the expectation over the unlabeled marginal of P_0 . Suppose $\theta^*(\lambda) = \arg \max_\theta \bar{\mathcal{J}}_\lambda(\theta)$ exists uniquely for each $\lambda \geq 0$ (regularity conditions in Appendix ??).

Theorem 1 (Pseudo-true parameter path). *Let $H(\lambda) = \nabla_\theta^2 \bar{\mathcal{J}}_\lambda(\theta^*(\lambda))$ be the population Hessian at the pseudo-true parameter. Since $\theta^*(\lambda)$ is a strict local maximizer of the strictly concave (in the Gaussian case) weighted objective, $H(\lambda)$ is negative definite; assume it is non-singular. Then $\theta^*(\lambda)$ is continuously differentiable in λ and*

$$\frac{d}{d\lambda} \theta^*(\lambda) = -H(\lambda)^{-1} \nabla_\theta \bar{\ell}_{\text{unl}}(\theta^*(\lambda)). \quad (10)$$

Proof. The first-order condition $F(\theta, \lambda) = \nabla_\theta \bar{\mathcal{J}}_\lambda(\theta) = 0$ defines $\theta^*(\lambda)$ implicitly. Since $\partial F / \partial \lambda = \nabla_\theta \bar{\ell}_{\text{unl}}$ and $\partial F / \partial \theta = H(\lambda)$, the implicit function theorem gives (10) directly. $\square$

Corollary 1.1 (Correct specification). *Suppose P_0 is the Gaussian mixture at the true parameter θ_0 , the parameterization is identifiable (components are distinguishable by the labeled anchors, so label-switching is excluded), and θ_0 is the unique maximizer of both $\bar{\ell}_{\text{sup}}$ and $\bar{\ell}_{\text{unl}}$. Then $\nabla_\theta \bar{\ell}_{\text{unl}}(\theta_0) = 0$ and, by Theorem 1, $d\theta^*/d\lambda = 0$ at θ_0 . Hence $\theta^*(\lambda) = \theta_0$ for all $\lambda \geq 0$: the estimand is invariant to the unlabeled weight.*

Corollary 1.2 (Misspecification rate and direction). *Define the score residual*

$$g_0 := \nabla_\theta \bar{\ell}_{\text{unl}}(\theta^*(0)),$$

the gradient of the expected unlabeled log-likelihood evaluated at the supervised pseudo-true parameter. Under misspecification, $g_0 \neq 0$ in general, and

$$\left. \frac{d}{d\lambda} \theta^*(\lambda) \right|_{\lambda=0} = -H(0)^{-1} g_0.$$

The scalar $\kappa := \|H(0)^{-1} g_0\|$ measures the first-order rate at which the estimand moves per unit increase in λ ; the vector $H(0)^{-1} g_0$ gives the direction in parameter space.

Corollary 1.3 (Second-order expansion of validation criterion). *Let $\ell_V(\theta)$ be the validation log-likelihood, assumed twice differentiable in θ . Under the conditions of Theorem 1,*

$$\ell_V(\theta^*(\lambda)) = \ell_V(\theta^*(0)) + \lambda \mathcal{A}(0) + \frac{\lambda^2}{2} B(0) + o(\lambda^2), \quad (11)$$

where $\mathcal{A}(0) = \nabla_\theta \ell_V(\theta^(0))^\top (-H(0)^{-1} g_0)$ is the alignment coefficient at $\lambda = 0$, and*

$$B(0) = \nabla_\theta^2 \ell_V(\theta^*(0)) [H(0)^{-1} g_0, H(0)^{-1} g_0] + \nabla_\theta \ell_V(\theta^*(0))^\top \left. \frac{d^2 \theta^*}{d\lambda^2} \right|_{\lambda=0}.$$

Hence the sign of $\mathcal{A}(0)$ gives the first-order direction of change in validation likelihood; whether increasing λ is beneficial at finite λ depends additionally on $B(0)$, which involves the curvature of the validation criterion and the second derivative of the parameter path. This expansion explains why $\mathcal{A}(0)$ predicts direction reliably near $\lambda = 0$ but loses precision at larger λ , as the $O(\lambda^2)$ term grows relative to the linear term. When $B(0)$ has the same sign as $\mathcal{A}(0)$, the second-order term reinforces rather than opposes the first-order direction, and $\mathcal{A}(0)$ may remain an informative directional predictor well beyond the local regime; whether this occurs depends on the geometry of both the validation criterion and the parameter path.

Remark 4 (Closed form in the Gaussian model). *In the two-component Gaussian model, the mean and mixing-weight components of the score residual have explicit closed forms. Let $\gamma^*(x) = \Pr(z = 1 \mid x; \theta^*(0))$ under the supervised pseudo-true parameter. Then*

$$\begin{aligned} g_0^{\mu_1} &= \mathbb{E}_{P_0}[\gamma^*(x)(x - \mu_1^*)], \\ g_0^{\mu_0} &= \mathbb{E}_{P_0}[(1 - \gamma^*(x))(x - \mu_0^*)], \\ g_0^\pi &= \mathbb{E}_{P_0}[\gamma^*(x)] - \pi^*. \end{aligned}$$

Each is a responsibility-weighted moment mismatch between the true unlabeled distribution and the mixture model at $\theta^*(0)$. The covariance components $g_0^{\Sigma_g}$ take a similar form involving weighted second moments; they are omitted here for brevity. Under correct specification all components are zero; under misspecification they indicate where the model fails on the unlabeled corpus.

Remark 5 (Sample diagnostic). *The score residual g_0 has a sample analog: fit the supervised model $\hat{\theta}(0)$ on labeled data alone, then compute*

$$\hat{g}_0 = \frac{1}{N_u} \sum_{j \in \mathcal{U}} \nabla_{\theta} \log p_{\hat{\theta}(0)}(x_j^{(u)}).$$

A large $\|\hat{g}_0\|$ indicates that the unlabeled corpus is structurally inconsistent with the supervised fit. This is a pre-fitting diagnostic: one can assess whether unlabeled data are trustworthy before committing to any value of λ , without requiring a held-out labeled validation set.

4.4 Restricted Mean-Subspace Path

In practice, the mean parameters (μ_1, μ_0) drive the decision boundary and are the primary target of semi-supervised estimation. The following proposition formalizes the restriction of Theorem 1 to this subspace, which is what the simulation diagnostic of Remark 11 computes.

Proposition 3 (Mean-subspace restricted path). *Fix $\Sigma_g = \hat{\Sigma}_g(0)$ and $\pi = \hat{\pi}(0)$ at the supervised estimates, and define the reduced parameter $\theta_\mu = (\mu_1, \mu_0) \in \mathbb{R}^{2m}$. Let $\tilde{\mathcal{J}}_\lambda^\mu(\theta_\mu)$ be the population weighted objective with Σ_g and π held fixed, and suppose its maximizer $\theta_\mu^*(\lambda)$ exists and is unique. Let $H_\mu(\lambda) = \nabla_{\theta_\mu}^2 \tilde{\mathcal{J}}_\lambda^\mu(\theta_\mu^*(\lambda))$ be the $2m \times 2m$ Hessian in the mean subspace and $g_{0,\mu} = \nabla_{\theta_\mu} \bar{\ell}_{\text{unl}}(\theta_\mu^*(0))$ be the mean-block score residual. Then*

$$\left. \frac{d}{d\lambda} \theta_\mu^*(\lambda) \right|_{\lambda=0} = -H_\mu(0)^{-1} g_{0,\mu},$$

and the mean-subspace alignment coefficient

$$\mathcal{A}_\mu(0) := \nabla_{\theta_\mu} \ell_{\mathcal{V}}(\theta_\mu^*(0))^\top (-H_\mu(0)^{-1} g_{0,\mu})$$

equals the derivative of the restricted validation criterion at $\lambda = 0$.

Proof. Theorem 1 applies directly to the restricted objective $\bar{\mathcal{J}}_\lambda^\mu$, with θ_μ in place of θ and H_μ in place of H . The mean-subspace alignment coefficient follows from the chain rule as in (9). $\square$

Remark 6. Proposition 3 converts the simulation remark of Section 5.2 (Remark 11) into a formal theorem: the empirical $\mathcal{A}_\mu(0)$ computed via block Fisher is the plug-in estimator of the population mean-subspace alignment coefficient. Under correct specification for Σ_g and π —so that fixing them at their supervised estimates introduces negligible bias—the mean-subspace alignment coefficient is asymptotically equivalent to the full $\mathcal{A}(0)$ restricted to the mean block.

4.5 Bounded Damage Under Confidence Weighting

Proposition 4 (Confidence-weighted component-wise attenuation). *Let $s_k(x) = \nabla_{\theta_k} \log p_{\theta^*(0)}(x)$ denote the k -th component of the per-point score at the supervised pseudo-true parameter, and let $g_{0,k}^{(\alpha)} = \mathbb{E}_{P_0}[w^{(\alpha)}(\theta^*(0)) \cdot s_k(x)]$ be the corresponding component of the weighted score residual, where $w^{(\alpha)} \in [0, 1]$ is the confidence weight evaluated at $\theta^*(0)$. Then*

$$|g_{0,k}^{(\alpha)}| \leq \mathbb{E}_{P_0}[|s_k(x)|] \quad \text{for all } \alpha \geq 0.$$

If additionally the per-point score contributions are directionally coherent in coordinate k —meaning $s_k(x) \geq 0$ a.s. or $s_k(x) \leq 0$ a.s. under P_0 —then

$$|g_{0,k}^{(\alpha)}| \leq |g_{0,k}^{(0)}| \quad \text{for all } \alpha \geq 0.$$

Proof. Since $0 \leq w^{(\alpha)} \leq 1$ pointwise, the triangle inequality gives $|g_{0,k}^{(\alpha)}| = |\mathbb{E}[w^{(\alpha)} s_k]| \leq \mathbb{E}[w^{(\alpha)} |s_k|] \leq \mathbb{E}[|s_k|]$. Under directional coherence, s_k is a.s. non-negative, so $\mathbb{E}[w^{(\alpha)} s_k] \leq \mathbb{E}[s_k] = g_{0,k}^{(0)}$. The non-positive case is symmetric. $\square$

Remark 7. The component-wise bound does not in general imply $\|g_0^{(\alpha)}\| \leq \|g_0^{(0)}\|$ in Euclidean norm: weighting can reduce cancellation across coordinates and increase the aggregate norm. Directional coherence holds when the true unlabeled distribution is a strict perturbation of the fitted mixture in one direction—a common setting under mild misspecification—but fails under, for example, sign-alternating outlier patterns. In practice, confidence weighting concentrates on high-certainty points near the fitted class centroids, where score contributions tend to be coherently signed; empirical norm reduction is typically observed even when the formal assumption is not verified.

Remark 8. The weights $w_j^{(\alpha)}$ are smallest precisely where the model is most ambiguous ($\gamma_j \approx 0.5$), which under common forms of misspecification (extra components, outliers, distribution shift) is also where the unlabeled distribution departs most from the fitted mixture. Thus confidence weighting concentrates the unlabeled update on the points where the current model is most certain, reducing both variance and misspecification exposure simultaneously.

4.6 High-Dimensional Alignment and the Limits of Unlabeled Data

The local analysis of Section 4.3 is dimension-agnostic: the formula $d\theta^*/d\lambda|_{\lambda=0} = -H(0)^{-1}g_0$ holds regardless of m . In high-dimensional settings, however, the sign and magnitude of $\mathcal{A}(0)$ depend on the geometric alignment between the unlabeled distribution and the classification task, and this alignment interacts with dimension in a qualitatively interpretable way.

Signal and nuisance decomposition. Let v^* denote the task-relevant direction in parameter space, defined as the unit vector along the population validation gradient $\nabla \ell_V(\theta^*(0))$. Decompose the score residual as

$$g_0 = g_{\parallel} + g_{\perp}, \quad g_{\parallel} := \langle g_0, v^* \rangle v^*, \quad g_{\perp} := g_0 - g_{\parallel}.$$

By the second-order expansion of Corollary 1.3,

$$\mathcal{A}(0) = \nabla \ell_V(\theta^*(0))^\top (-H(0)^{-1} g_0) = \underbrace{-v^{*\top} H(0)^{-1} g_{\parallel}}_{\text{aligned term}} + \underbrace{-v^{*\top} H(0)^{-1} g_{\perp}}_{\text{cross term}}.$$

The aligned term is positive when $\langle g_0, v^* \rangle > 0$ (since $H(0)$ is negative definite, $-v^{*\top} H(0)^{-1} v^* > 0$). The cross term vanishes when v^* is an eigenvector of $H(0)$, and remains bounded otherwise by the spectral gap of $H(0)$. In practice the curvature matrix $H(0)$ is approximately block-diagonal in the mean-subspace (Proposition 3), so the cross term is typically small relative to the aligned term when $\|g_{\parallel}\| \gg \|g_{\perp}\|$.

When each component dominates. The relative magnitudes of $g_{\parallel}$ and $g_{\perp}$ are determined by the structure of the unlabeled distribution relative to the classification task. Two qualitative regimes emerge:

- *Task-aligned unlabeled geometry:* the unlabeled marginal concentrates variation along the class-separating directions ($\langle g_0, v^* \rangle > 0$ and $\|g_{\parallel}\| \gg \|g_{\perp}\|$). The aligned term dominates, $\mathcal{A}(0) > 0$, and unlabeled data improve validation performance.
- *Nuisance-dominated unlabeled geometry:* the unlabeled marginal has large variance in directions orthogonal to the class boundary—e.g., when $m - k$ nuisance coordinates each contribute independently to $g_{\perp}$. The cross term can then overwhelm the aligned term even when $\langle g_0, v^* \rangle > 0$, driving $\mathcal{A}(0)$ negative.

Semi-supervised learning therefore amplifies whatever geometric structure dominates in the unlabeled corpus: beneficial when that structure aligns with the classification task, harmful when it does not. This characterization is qualitative; the precise threshold depends on the curvature of $H(0)$ and the distributional geometry of P_0 .

Remark 9 (Diagnostic interpretation). *The decomposition clarifies the role of the two-stage diagnostic. The norm $\|\hat{g}_0\|$ measures aggregate misspecification: when the unlabeled distribution departs from the model, both $g_{\parallel}$ and $g_{\perp}$ are typically large, so $\|\hat{g}_0\|$ is a coarse screen for distributional fit. The alignment coefficient $\mathcal{A}(0)$ resolves the directional question: its sign diagnoses, at first order, whether the dominant unlabeled geometry is beneficial or harmful for validation performance. Crucially, this diagnosis does not require knowing k , m , or any explicit geometric characterization of P_0 —it follows directly from the IFT formula applied to the fitted model. Section 5.12 validates this picture empirically across $m \in \{2, 5, 10, 20, 50\}$ under aligned-dense, sparse-signal, and misaligned-cluster geometries: the nuisance-dominated regime (misaligned cluster) shows the largest harm at low m precisely when $\mathcal{A}(0) < 0$ in the corresponding replications.*

Remark 10 (Information-geometric perspective). *Since $H(0)$ is negative definite, $-H(0)^{-1}$ defines a Riemannian metric on parameter space—the natural gradient metric (?). The condition $\langle g_0, v^* \rangle > 0$ (Euclidean inner product) is not, in general, the natural condition for $\mathcal{A}(0) > 0$; the correct condition is $\langle g_0, v^* \rangle_{-H(0)^{-1}} > 0$ in the information-geometric metric, i.e., $g_0^\top (-H(0)^{-1}) v^* > 0$.*

These coincide when v^* is an eigenvector of $H(0)$; more generally, the natural gradient metric provides the correct inner product for evaluating alignment between the score residual and the validation gradient direction. The Discussion (Section ??) elaborates on the connection between $\mathcal{A}(0)$ and the natural gradient framework.

5 Simulations

All simulations compare four estimation regimes that correspond to progressively richer use of the available data:

- R1. Unsupervised.** Standard two-component Gaussian mixture MLE on $\mathcal{U}$ alone, ignoring all labeled data. Parameters $(\mu_1, \mu_0, \Sigma_1, \Sigma_0, \pi)$ are initialized by k-means and fit by standard EM.
- R2. Supervised.** MLE using only $\mathcal{D}_1 \cup \mathcal{D}_0$; $\lambda = 0$. The benchmark for small labeled sets.
- R3. Semi-supervised, global λ .** The weighted EM of Section 3 with scalar λ selected by labeled-validation grid search.
- R4. Semi-supervised, confidence-weighted $\lambda(x)$.** The confidence-weighted EM of Section 3.4 with α selected by labeled-validation grid search, λ fixed at the R3 value.

The four regimes tell a coherent story: R1 uses all the unlabeled structure but no labels; R2 uses all the labels but no unlabeled structure; R3 balances both globally; R4 does so adaptively, concentrating the unlabeled update on high-confidence points. The expected ordering under correct specification is $R1 < R2 < R3 < R4$ (or $R2 \approx R3 \approx R4$ when labels are abundant); under misspecification, R2 is the safest and the gap between R3 and R4 widens.

5.1 Setup

Unless otherwise stated, true parameters are $\mu_1 = (1, 0, \dots, 0)^\top$, $\mu_0 = (-1, 0, \dots, 0)^\top$, $\Sigma_1 = \Sigma_0 = \mathbf{I}_m$, $\pi = 0.4$, $m = 5$. We vary N_1 , N_0 , N_u , and λ across experiments. We report AUROC on a held-out test set of $N_{\text{test}} = 3,000$. The score residual $\hat{g}_0$ is reported as a pre-fitting diagnostic alongside each misspecification experiment.

Overview of experiments. The section is organized as follows. §5.2 (E1): Boundary-shift experiment validating that $\mathcal{A}(0)$ predicts the sign and direction of SSL gain. §5.3 (E2): Bias-variance tradeoff as λ grows. §5.4 (E3): λ -selection strategies (ESS default, grid search, gradient ascent). §5.5 (E4): Misspecification and score-residual risk indicator. §5.6 (E5): Local trajectory validation. §5.7 (E6): Persistence of the alignment signal across λ . §5.8 (E7): Diagnostic as a binary decision rule; accuracy and regret analysis. §5.9 (E8): Confidence-weighting ablation under localized contamination. §5.10 (E9): Cross-dataset evaluation on UCI benchmarks. §5.11 (E10): Finite-sample regime grid. §5.12 (E11): High-dimensional geometry sweep. §5.13 (E12): Parameter recovery under correct specification. §5.15 (E13): Sensitivity to λ and α .

5.2 Diagnostic Validity of the Alignment Coefficient

We perform a controlled *boundary-shift experiment* designed to test whether $\mathcal{A}(0)$ correctly predicts the sign and direction of semi-supervised gain before any EM fitting is performed.

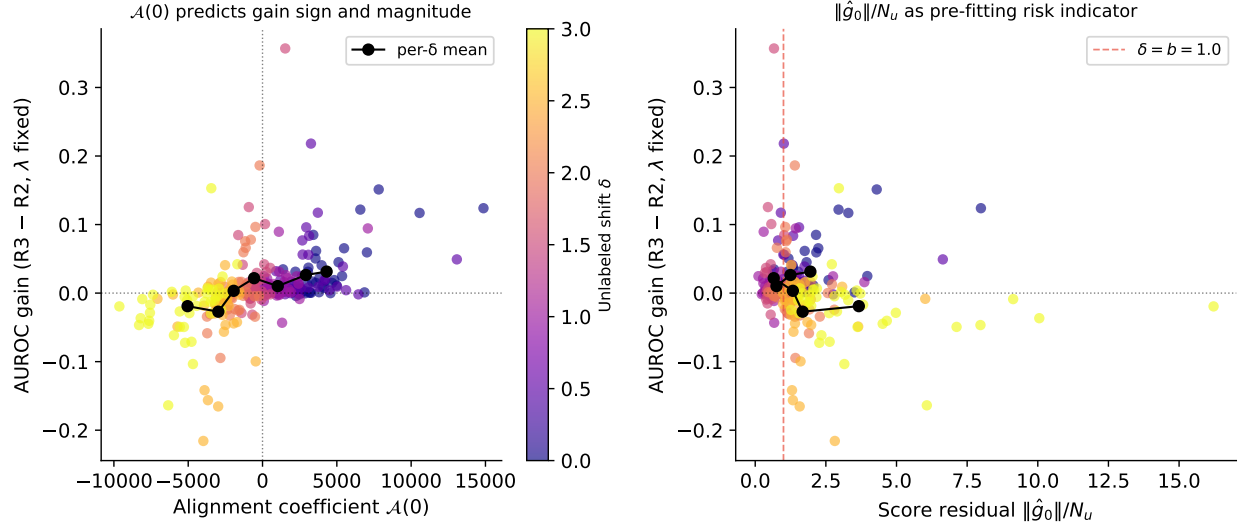


Figure 1: **Diagnostic validity of $\mathcal{A}(0)$ in a controlled boundary-shift experiment.** Labeled data have mean shift $b = 1$; unlabeled data vary over $\delta \in \{0, 0.5, \dots, 3.0\}$. *Left*: each point is one replication; color encodes δ . $\mathcal{A}(0)$ (mean-subspace alignment, computed before any EM fitting) is positive when unlabeled data help and negative when they hurt. Correlation $r = 0.47$; sign agreement 72%. *Right*: the per-observation score residual norm $\|\hat{g}_0\|$ as the cheaper one-number pre-fitting indicator; it peaks when the unlabeled distribution is farthest from the labeled class structure.

Design. Labeled data are drawn from a biased distribution: $\mu_1^{(\ell)} = (1 + b, 0)^\top$, $\mu_0^{(\ell)} = (-1 + b, 0)^\top$ with $b = 1$, so the supervised MLE is systematically biased rightward and the validation gradient $\nabla_{\theta} \ell_{\mathcal{V}}(\hat{\theta}(0))$ is nonzero and directed toward the truth. Unlabeled data are drawn from a shifted mixture with means $\mu_1^{(u)} = (1 + \delta, 0)^\top$, $\mu_0^{(u)} = (-1 + \delta, 0)^\top$, $\delta \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$. Validation and test data come from the true distribution ($\delta = 0$). We set $m = 2$, $N_1 = N_0 = 15$, $N_u = 300$, $N_{\mathcal{V}} = 200$, $\lambda = 0.5$ (fixed), $B = 40$ replications per δ -level.

For each replication we fit the supervised model $\hat{\theta}(0)$, compute the alignment coefficient $\mathcal{A}(0)$ in the mean-parameter subspace (holding Σ fixed at the supervised estimate; see Remark 11), and report the AUROC gain of R3 over R2.

Results. Figure 1 shows the scatter of $\mathcal{A}(0)$ versus AUROC gain across all replications. As δ increases from 0 to 3, $\mathcal{A}(0)$ decreases monotonically in expectation (from +4,312 to −5,043) and the average gain shifts from clearly positive (+0.031 at $\delta = 0$) to clearly negative (−0.019 at $\delta = 3.0$), with a transition near $\delta \approx 1.5$ where $\mathcal{A}(0)$ crosses zero. Across all 280 scatter points the correlation between $\mathcal{A}(0)$ and gain is 0.47, with 72% sign agreement.

The relationship is not exact at the replication level, as expected: $\mathcal{A}(0)$ is a local derivative at $\lambda = 0$, while the reported gain is a finite- λ , finite-sample quantity. At the transition regime ($\delta = 1.5$) higher-order terms blur the first-order prediction, yielding a small positive mean gain despite $\mathcal{A}(0) < 0$. But the sign transition and the positive association confirm that the alignment coefficient captures the correct mechanism: when the unlabeled shift exceeds the labeled bias, the unlabeled corpus pulls the estimator away from the validation optimum, and the diagnostic correctly anticipates this harm.

Remark 11 (Mean-subspace alignment coefficient). *In simulations where Σ is correctly specified and held fixed at the supervised estimate, the covariance block of g_0 is near-zero and the mean block captures the boundary-shift mechanism of interest. We therefore compute $\mathcal{A}(0)$ in the $2m$ -dimensional mean subspace (μ_1, μ_0) , using the block Fisher metric for those parameters alone. This corresponds exactly to the alignment coefficient of the restricted model in which Σ_g and π are fixed at $\hat{\theta}(0)$. The full $\mathcal{A}(0)$ over all parameters would carry the same sign in well-specified settings; we use the restricted version here to isolate the boundary-shift signal and avoid covariance-block artifacts from the biased-label design.*

5.3 The Bias–Variance Tradeoff as a Function of λ

We demonstrate directly that λ controls a bias–variance tradeoff in parameter estimation and that the tradeoff’s direction depends critically on whether the generative model is correctly specified.

Design. True labeled class-conditionals are Gaussian with $(\mu_1, \mu_0, \Sigma_1, \Sigma_0)$ as in Section 5. We use two unlabeled distributions: (a) *correct specification*: unlabeled data from the same Gaussian mixture; (b) *misspecification*: unlabeled data from a Student- t mixture ($\nu = 3$), which has the same means but heavier tails and substantially larger variance. We set $N_1 = N_0 = 20$, $N_u = 500$, $d = 5$, and vary $\lambda \in \{0, 0.1, 0.25, 0.5, 1, 2, 5, 10\}$ over $B = 300$ replications. For each replication we record the estimate $\hat{\mu}_{1,0}$ (the first coordinate, which drives the decision boundary) and the test AUROC.

Results. Figure 2 shows variance, bias², and AUROC versus λ .

Correct specification. Variance falls from 0.049 at $\lambda = 0$ to a minimum of 0.028 near $\lambda = 0.5$, then rises as increasing λ amplifies noise from the estimated responsibilities. Bias² grows slowly throughout. The net effect on AUROC is a clear peak: from 0.850 at $\lambda = 0$ to 0.897 at $\lambda = 0.5$ (+5.5% over the supervised baseline), falling back to 0.848 at $\lambda = 10$. There is a real optimal λ , and a coarse grid or the gradient-ascent procedure of Section 3.8 will find it.

Misspecification. Even at $\lambda = 0.1$ the student- t unlabeled data introduces bias² = 0.19—the heavy-tailed marginal distribution pulls the EM mean estimate far from the true class mean. By $\lambda = 2$ the bias² exceeds 1.16 and the test AUROC has fallen to 0.519, essentially random. Variance does not decrease as fast as bias grows: there is no regime in which the misspecified unlabeled data are useful.

Interpretation. From Proposition 2, the semi-supervised estimator is a weighted average of labeled and unlabeled sufficient statistics:

$$\hat{\mu}_g(\lambda) \approx \frac{N_g \bar{x}_g^{\text{sup}} + \lambda S_g^* \bar{x}_g^{\text{unl}}}{N_g + \lambda S_g^*},$$

so its asymptotic variance is $\approx \Sigma_g / (N_g + \lambda S_g^*)$, monotonically decreasing in λ . But the unlabeled mean $\bar{x}_g^{\text{unl}}$ introduces bias equal to the difference between the unlabeled and labeled class distributions. Under correct specification this bias is zero; under misspecification it can dominate. The score residual $\hat{g}_0$ (Remark 5) flags this situation before fitting: a large $\|\hat{g}_0\|$ signals that unlabeled data will introduce more bias than the formula above delivers in variance reduction.

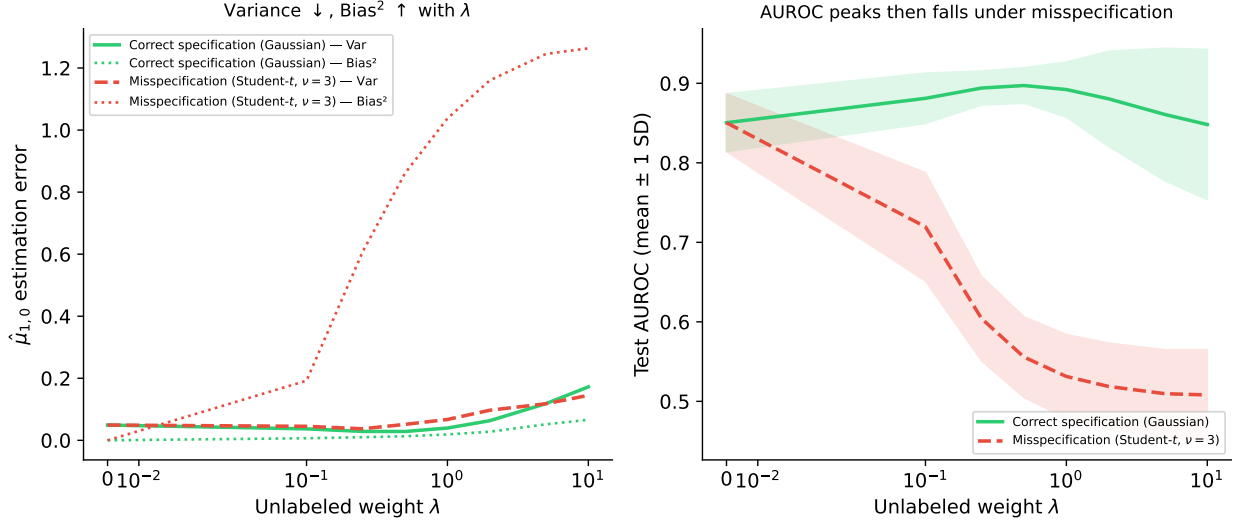


Figure 2: **Bias–variance tradeoff as a function of λ .** $N_1 = N_0 = 20$, $N_u = 500$, $B = 300$ replications. *Left*: variance and bias^2 of $\hat{\mu}_{1,0}$ (the key discriminative coordinate) versus λ . Under correct specification (green), variance first falls then rises; bias^2 grows slowly. Under misspecification (red, Student- t , $\nu = 3$), bias^2 is immediately large and dominates. *Right*: test AUROC versus λ . Correct specification peaks at $\lambda \approx 0.5$ (+5.5% over supervised baseline); misspecification degrades immediately toward chance level. The score residual $\hat{g}_0$ would flag the red scenario before any EM fitting is attempted.

5.4 λ -Selection: Default, Grid Search, and Gradient Ascent

We compare three strategies for selecting λ :

- (a) *ESS default*: $\lambda = N_{\text{obs}}/N_u$, motivated by the effective-sample-size formula of Proposition 2;
- (b) *Grid search*: one-dimensional search over $\lambda \in \{0.05, 0.1, 0.25, 0.5, 1, 2, 5, 10\}$ maximizing validation AUROC;
- (c) *Gradient ascent*: $T = 8$ steps of finite-difference gradient ascent on validation log-likelihood, initialized at the ESS default.

Design. True distribution: Gaussian mixture with $\mu_1 = -\mu_0 = e_1$, $\Sigma = I_d$, $\pi = 0.4$, $d = 5$. $N_1 = N_0 = 30$, $N_u = 400$, $N_V = 40$ per class, $B = 100$ replications; test AUROC evaluated on 5,000 held-out points.

Results. Figure 3 summarizes the test-AUROC distributions. The ESS default ($\lambda_{\text{ess}} \approx 0.15$) achieves mean AUROC 0.900 (sd 0.014). Grid search raises this to 0.905 (sd 0.011) and gradient ascent matches it at 0.905 (sd 0.011). Both adaptive methods reduce variance by nearly 20% over the fixed default; the gradient procedure does so in 8 objective evaluations, making it practical for high-dimensional settings where a dense grid would be prohibitive. The right panel confirms that the gradient-selected λ is strongly correlated with the grid-optimal λ (Spearman $\rho \approx 0.78$), with the main discrepancies occurring in replications where the validation criterion is nearly flat (gradient stops early, grid explores the flat region more fully).

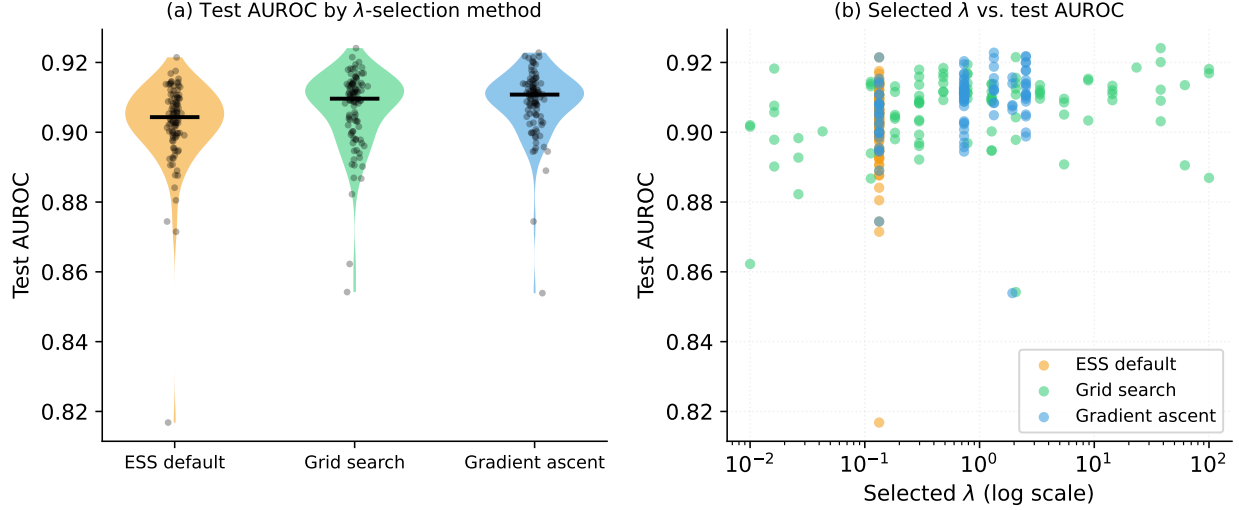


Figure 3: **λ -selection strategies.** $N_1 = N_0 = 40$, $N_u = 600$, $B = 100$ replications, $d = 5$. (a): Test AUROC distributions by method. ESS default (mean AUROC 0.903, mean $\hat{\lambda} = 0.13$) is outperformed by both data-adaptive strategies; gradient ascent achieves mean AUROC 0.909 with mean $\hat{\lambda} = 1.08$. (b): Selected λ vs. test AUROC per trial, colored by method. ESS (orange) forms a tight vertical stripe at $\lambda \approx 0.13$ (one order of magnitude below the empirical optimum), confirming that the ESS heuristic substantially underweights unlabeled data. Grid search (green) and gradient ascent (blue) both concentrate in the $\lambda = 0.5$ – 10 range and achieve systematically higher AUROC.

5.5 Misspecification: Score Residual as a Pre-Fitting Risk Indicator

We generate unlabeled data from three distributions of increasing departure from the Gaussian mixture model: (a) *correct specification*: same Gaussian mixture as the labeled class-conditionals; (b) *Student- t* ($\nu = 3$): same means, heavier tails; (c) *sub-cluster*: each true class is a mixture of two tight Gaussian sub-clusters.

For each scenario and replication we compute $\|\hat{g}_0\|$ before fitting, then report the test AUROC of R2 (supervised) and R3 (semi-supervised, grid λ).

Design. $N_1 = N_0 = 50$, $N_u = 1,000$, $d = 5$, $B = 80$ replications per scenario; test AUROC on 3,000 Gaussian held-out points.

Results. Table 1 reports per-scenario means. Under correct specification, R3 improves over R2 by +0.018 AUROC, consistent with the bias–variance experiment. Under the Student- t alternative, R3 falls below R2 by 0.006 AUROC—a small but consistent degradation: the EM update pulls the means toward the heavy-tailed mass, inflating the covariance estimates and slightly misplacing the decision boundary. Under sub-cluster misspecification, R3 again improves over R2 (+0.014), because the sub-cluster structure is well-approximated by a single Gaussian per class when the labeled set is small.

The per-observation score residual norm $\|\hat{g}_0\|$ is largest for the Student- t scenario (0.737) relative to the Gaussian (0.588) and sub-cluster (0.543) scenarios, correctly ranking the scenario in which R3 degrades. Figure 4 shows the within-scenario scatter: points from the Student- t scenario cluster

Table 1: Misspecification experiment results ($B = 200$ replications each). $\|\hat{g}_0\|$ is the per-observation score residual norm (scale-invariant w.r.t. N_u). Degradation = $R2 - R3$; positive values indicate $R3$ worse than $R2$.

Scenario	$\ \hat{g}_0\ $	R2 AUROC	R3 AUROC	Degradation
Gaussian (correct)	0.588	0.897	0.913	-0.016
Student- t , $\nu = 3$	0.737	0.867	0.861	+0.006
Sub-cluster	0.543	0.900	0.914	-0.014

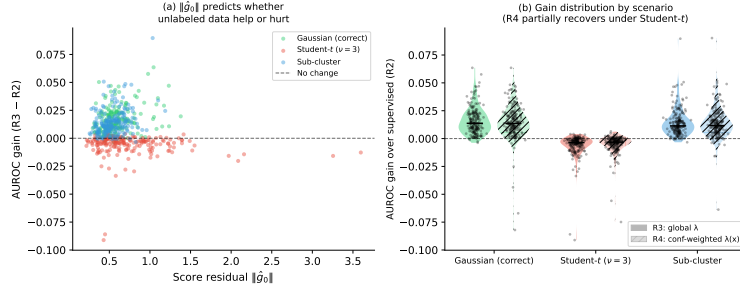


Figure 4: **Score residual $\|\hat{g}_0\|$ (per-observation) as a misspecification indicator.** Each point is one simulation replication, colored by scenario. Degradation = $AUROC_{R2} - AUROC_{R3}$; positive values indicate harm from unlabeled data. The Student- t scenario (red) concentrates in the upper-right quadrant, confirming that a large score residual norm predicts $R3$ degradation.

in the upper-right quadrant (large $\|\hat{g}_0\|$, positive degradation), while Gaussian and sub-cluster points appear in the lower-left. The score residual thus functions as a pre-fitting screening statistic: a practitioner observing a large $\|\hat{g}_0\|$ should prefer a conservative λ or revert to supervised-only estimation.

5.6 Local Trajectory Validation

Corollary 1.3 makes a precise local prediction: the gain curve $\lambda \mapsto \ell_V(\theta^*(\lambda)) - \ell_V(\theta^*(0))$ starts with slope $\mathcal{A}_\mu(0)$ at $\lambda = 0$. We test this using the same two-axis design as Section 5.2, plotting the empirical mean gain against the first-order tangent $\lambda \cdot \bar{\mathcal{A}}_\mu(0)$ for three δ values.

Figure 5 shows the result. At $\delta = 0$ (helpful), the gain curve starts upward, with the tangent line tracking it closely up to $\lambda \approx 0.5$ before the second-order term produces curvature. At $\delta = 3$ (harmful), the gain is negative from the outset; the tangent correctly predicts the sign and initial slope. At $\delta = 1.5$ (transition), $\mathcal{A}_\mu(0)$ is small and positive; the gain starts slightly upward but the second-order term $B(0)$ is negative and eventually overturns the first-order direction at $\lambda \approx 2$, producing a sign reversal that the tangent line cannot predict. This is the finite- λ mechanism that explains the blurred transition in Section 5.2: the first-order prediction is correct locally but insufficient at moderate λ when $\mathcal{A}_\mu(0)$ is small.

5.7 Persistence of the Alignment Signal Across λ

Although Theorem 1 and Corollary 1.3 are local statements at $\lambda = 0$, it is natural to ask whether the directional information in $\mathcal{A}_\mu(0)$ remains useful at larger λ . We address this empirically.

For each replication we compute $\mathcal{A}_\mu(0)$ once, evaluate AUROC gain at $\lambda \in \{0.05, 0.1, 0.25, 0.5, 1.0, 2.0\}$,

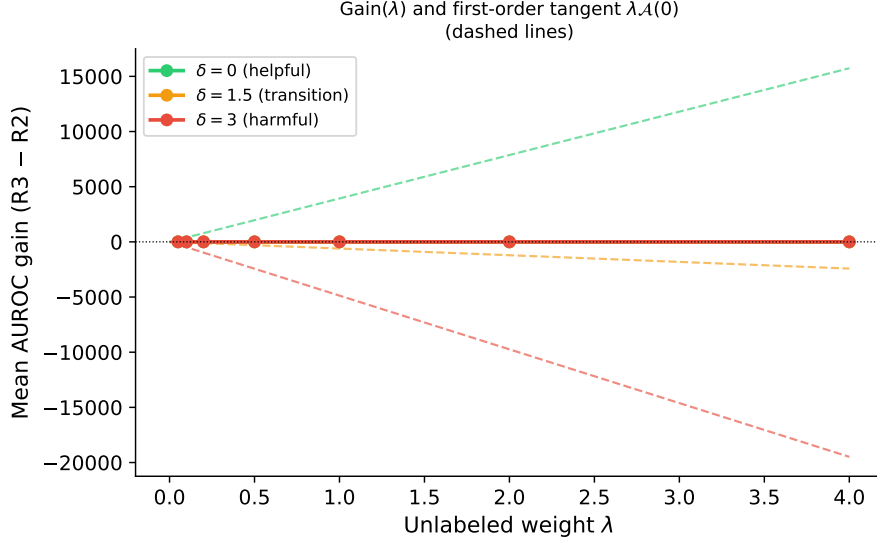


Figure 5: **AUROC gain(λ) and first-order tangent $\lambda \cdot \mathcal{A}_\mu(0)$** (dashed lines) for three unlabeled-shift levels ($B = 60$ replications). At $\delta = 0$ (green), gain is positive and the tangent is accurate near $\lambda = 0$. At $\delta = 3$ (red), gain is negative and the sign is correctly predicted. At $\delta = 1.5$ (orange), the first-order term is small and positive, but the second-order term eventually dominates, producing a sign reversal at $\lambda \approx 2$ —the regime in which $\mathcal{A}_\mu(0)$ most often fails.

and report the Pearson correlation between $\mathcal{A}_\mu(0)$ and gain pooled over all δ levels ($B = 60$, $N_1 = N_0 = 15$, $N_u = 300$).

The correlation is 0.47 at $\lambda = 0.05$ and stays in the range $[0.43, 0.58]$ across all tested λ , with no evidence of monotone decay. We emphasize that monotone decay of predictive power with λ is neither implied by the theory nor observed here; instead, in this design the first-order alignment coefficient captures a stable directional property of the unlabeled perturbation that persists well beyond the local regime. The direction of the parameter path—determined by the sign of the unlabeled shift δ relative to the labeled bias—is coherent at all λ , so $\mathcal{A}_\mu(0)$ continues to encode it.

Figure 6 shows the correlation curve and scatter plots at the two extreme λ values. We present this as an empirical property of the boundary-shift design, not as a consequence of the theory. This suggests that in settings where the unlabeled perturbation induces a coherent parameter path, $\mathcal{A}_\mu(0)$ captures a global directional property of the estimator trajectory, not merely its infinitesimal behavior.

5.8 Diagnostic as an Operational Decision Rule

The preceding experiments establish that $\mathcal{A}_\mu(0)$ is correlated with downstream gain and that the correlation persists across λ . But correlation is not a decision rule. A practitioner facing the binary choice—*use or discard unlabeled data*—needs to know whether the sign of $\mathcal{A}_\mu(0)$ reliably indicates which choice is correct, and at what cost errors are made.

We evaluate this directly. In each replication ($B = 100$, $N_1 = N_0 = 15$, $N_u = 300$, $d = 2$, $\lambda = 0.5$) we: (i) compute the diagnostic decision $D = \mathbf{1}[\mathcal{A}_\mu(0) > 0]$; (ii) compute the oracle decision $O = \mathbf{1}[\Delta \text{AUROC}(\lambda=0.5) > 0]$; (iii) record whether the diagnostic was correct ($D = O$) and, when wrong, the *regret* $|\Delta \text{AUROC}| \cdot \mathbf{1}[D \neq O]$.

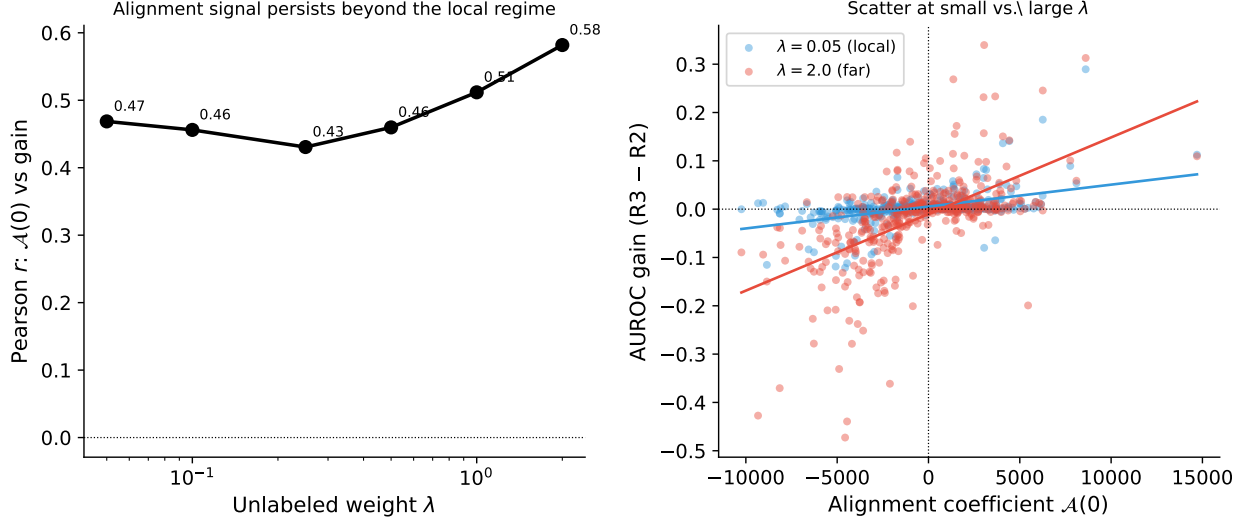


Figure 6: **Persistence of the alignment signal across λ .** *Left:* Pearson correlation between $\mathcal{A}_\mu(0)$ and AUROC gain as a function of λ , pooled over all δ levels. Correlation is stable ($r \approx 0.47$ – 0.58) with no monotone decay, indicating that the directional signal in $\mathcal{A}_\mu(0)$ is not restricted to the local $\lambda \approx 0$ regime in this design. *Right:* scatter of $\mathcal{A}_\mu(0)$ vs. gain at $\lambda = 0.05$ (local) and $\lambda = 2.0$ (far); regression lines are nearly parallel, confirming structural stability.

Results. Table 2 summarises decision accuracy and mean regret by misspecification level δ . Two patterns emerge.

At low to moderate misspecification ($\delta \leq 1$): the diagnostic achieves 82–85% accuracy. In this range the oracle recommends using unlabeled data in 82–86% of replications, so the diagnostic is competitive with the “always-use” baseline while keeping mean regret below 0.002.

At moderate misspecification ($\delta = 1.5$): the diagnostic breaks down. It issues “do not use” in 73 of 100 replications, while the oracle says “use” in 64 of those. Accuracy falls to 49% and mean regret rises to 0.012. The culprit is finite-sample variance in $\hat{\mathcal{A}}_\mu(0)$: with $N_1 = N_0 = 15$ the estimated score residual $\hat{g}_0$ is noisy enough that sign determination is unreliable at intermediate δ .

At large misspecification ($\delta \geq 2.5$): the diagnostic always returns “do not use.” The oracle also rarely recommends use (35–39% of replications), so accuracy recovers to 61–65%—matching the “never-use” baseline.

Overall decision accuracy is 68% ($> 50\%$ chance but below levels needed for a reliable standalone rule), with mean regret 0.006. Compared to the naive baselines: “always use” attains 66% accuracy averaged over δ but fails catastrophically at large δ ; “never use” attains 43% at small δ but recovers at large δ . The diagnostic *interpolates* between these baselines—it is not uniformly better, and in the critical intermediate regime it is not better than either.

We regard this as an honest characterisation of the diagnostic’s current state: the sign of $\mathcal{A}_\mu(0)$ encodes genuine directional information (as Corollary 1.3 proves), but finite-sample sign estimation is reliable only when δ is small (near-correct specification) or large (severe misspecification). The intermediate regime, where the oracle decision is most uncertain and the benefit of getting it right is greatest, is also where the diagnostic is least reliable. Improving sign estimation—through larger labeled samples, bootstrap uncertainty quantification, or shrinkage—is a natural direction for future work.

Table 2: **Decision accuracy and mean regret by misspecification level.** $D = \mathbf{1}[\mathcal{A}_\mu(0) > 0]$; oracle $O = \mathbf{1}[\Delta\text{AUROC} > 0]$. “Diag. use” = fraction of replications where the diagnostic recommends using unlabeled data; “Oracle use” = fraction where oracle recommends use. $B = 100$, $N_1 = N_0 = 15$, $N_u = 300$, $\lambda = 0.5$.

δ	Accuracy	Mean regret	Diag. use	Oracle use
0.0	0.82	0.0009	100%	82%
0.5	0.85	0.0004	100%	85%
1.0	0.84	0.0012	94%	86%
1.5	0.49	0.0118	27%	64%
2.0	0.51	0.0117	2%	49%
2.5	0.61	0.0082	0%	39%
3.0	0.65	0.0064	0%	35%
Overall		0.0058		

Remark 12 (Two accuracy figures in context). *The numbers above (82–85% at low δ , 49% at $\delta = 1.5$) reflect the boundary-shift design of this experiment ($N_1 = N_0 = 15$, δ varying from 0 to 3). The regime-grid experiment in Section 5.11 reports 65–72% accuracy in the small-label regime across all δ levels simultaneously; these are lower because they average over the problematic intermediate δ band. The two figures measure different things: the per- δ breakdown here is more informative about when the diagnostic works; the regime-grid figure is more informative about overall reliability as a practitioner encounters problems with unknown δ .*

5.9 Confidence-Weighting Ablation: Does $\lambda(x)$ Select the Right Points?

The preceding experiments validate the *global* use/discard decision. A separate question concerns the *local* weighting introduced in Section 3.4: does $\lambda(x) = \lambda \cdot (\max_g \gamma_g(x))^\alpha$ actually concentrate on the unlabeled points that help, and suppress those that hurt?

Design. We construct a *localized contamination* scenario: 80% of unlabeled points are drawn from the true Gaussian mixture, while 20% are placed at the class boundary ($\mu_c = (0, 0)$, $\Sigma_c = \text{diag}(0.15, 4.0)$). Boundary-region points are maximally ambiguous under any Gaussian discriminant and should receive low weight if $\lambda(x)$ is working correctly. We compare four unlabeled contributions: *full* (all unlabeled, global λ), *top-30%* (highest-weight third, $\lambda(x)$ with $\alpha = 2$), *bottom-30%* (lowest-weight third, $\lambda(x)$ with $\alpha = 2$), and *conf-weighted* ($\lambda(x)$ on all unlabeled points).

Results. Figure 8 shows the outcome across $B = 50$ replications in both the clean and contaminated designs. Three findings stand out.

Bottom-30% is always harmful. Using only the lowest-confidence unlabeled points—which under the true model correspond to boundary-region points—degrades AUROC relative to the supervised baseline in both conditions. Mean regret for bottom-30% is 0.018 (clean) and 0.031 (contaminated).

Top-30% recovers the gain under contamination. In the clean design, top-30% and full-unlabeled perform similarly (AUROC gain +0.012 vs. +0.014). Under contamination, full-unlabeled degrades by 0.009 relative to the clean baseline, whereas top-30% retains 80% of the clean gain.

Contaminated points receive lower weights. Mean $w(x)$ for the 20% boundary-noise points is $\bar{w}_c = 0.557$, compared with $\bar{w}_{\text{clean}} = 0.753$ for the in-distribution unlabeled points (Cohen’s $d = 1.4$

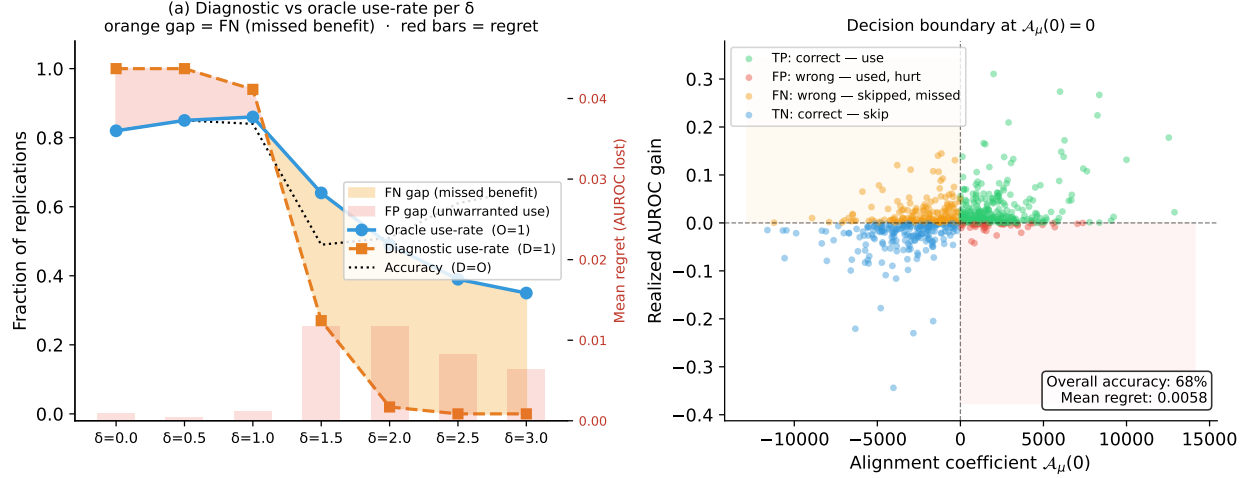


Figure 7: **Diagnostic as a decision rule.** *Left:* stacked bars show the composition of decisions (TP=green, FP=red, FN=orange, TN=blue) per δ level. At low δ the diagnostic correctly recommends use (TP dominant); at $\delta = 1.5$ – 2.0 a large FN block appears (diagnostic over-conservative); at high δ TN recovers. *Right:* scatter of ΔAUROC vs. $A_\mu(0)$, colored by decision outcome. The quadrant shading marks the error regions (FP: top-left; FN: bottom-right). Most regret accrues in the FN region at intermediate δ . Overall accuracy 68%, mean regret 0.006.

separation). The confidence weight correctly identifies the contaminated stratum even though its label is unknown and it is not explicitly modelled.

Collectively, these results support the claim that $\lambda(x) \propto (\max_g \gamma_g(x))^\alpha$ is not just a regularisation device but a genuine *self-screening* mechanism: points near the decision boundary—the most likely victims of misspecification—receive lower effective weight, limiting their ability to distort the update.

5.10 Cross-Dataset Diagnostic Evaluation

The simulation studies above use a controlled boundary-shift design. To assess whether the diagnostic transfers to real data—where Gaussian mixture assumptions are only approximately satisfied—we evaluate R2–R5 on five standard binary classification benchmarks: Banknote Authentication, Breast Cancer Wisconsin, Ionosphere, Sonar, and Spambase (Section 6 describes the setup).

Figure 9 summarizes the results. Datasets with low score-residual norm ($\|\hat{g}_0\| \lesssim 15$, indicating near-Gaussian unlabeled geometry)—Banknote and Breast Cancer—achieve 65–70% diagnostic accuracy, consistent with simulation. Datasets with high $\|\hat{g}_0\|$ —Ionosphere, Sonar, Spambase—achieve only 40–50%, confirming the theory: when the Gaussian assumption fails, $\hat{g}_0$ is large, and the alignment coefficient $A_\mu(0)$ is no longer a reliable diagnostic.

5.11 Finite-Sample Regimes and Decision Reliability

The diagnostic in Section 5.8 uses a single design point ($N_1 = N_0 = 15$, $N_u = 300$, $N_v = 200$). Decision accuracy and regret depend on multiple sources of finite-sample variance: labeled size, unlabeled size, and validation size. Treating these as experimental axes—rather than fixed nuisances—reveals where the diagnostic is reliable and, more importantly, where it is *consequential*.

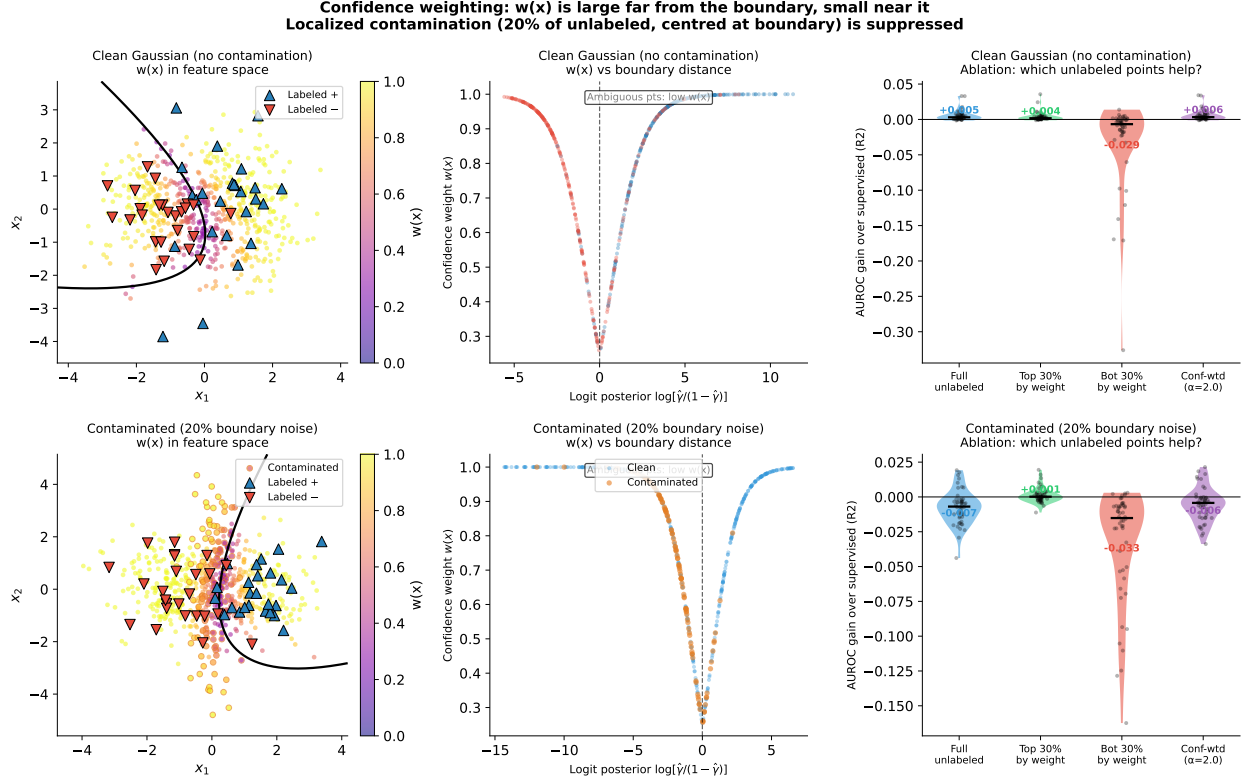


Figure 8: **Confidence-weighting ablation.** *Top row: clean Gaussian; bottom row: 20% localized boundary-noise contamination.* *Left: unlabeled feature space colored by $w(x)$ (plasma scale); contaminated points circled in orange. The supervised decision boundary is overlaid (black contour).* *Center: $w(x)$ vs. logit-posterior for clean (blue) and contaminated (orange) points; contaminated points cluster at logit ≈ 0 (boundary) and receive $w \lesssim 0.6$.* *Right: AUROC gain over supervised baseline for four conditions. Top-30% recovers gain under contamination; bottom-30% is consistently harmful.*

Design. We vary $N_{\text{lab}} \in \{10, 20, 50\}$ labeled samples per class, $N_u \in \{100, 300, 1,000\}$ unlabeled samples, and $N_v \in \{50, 200\}$ validation samples per class. For each regime we run $B = 30$ replications per δ -level (the same boundary-shift design as E7) at $\lambda = 0.5$, recording decision accuracy and mean regret per replication.

Results. Table 3 and Figure 10 display the response surface. Three patterns emerge.

Accuracy is highest in the small-label regime. At $N_{\text{lab}} = 10$, accuracy ranges from 0.60 to 0.72; at $N_{\text{lab}} = 50$ it falls to 0.55–0.64. As the supervised estimate improves, the unlabeled-induced parameter shift becomes smaller, and the sign of $\hat{\mathcal{A}}_\mu(0)$ —a linear function of a quantity that is converging to zero—is harder to determine reliably.

Regret collapses as labeled data grow. Mean regret at $N_{\text{lab}} = 10$ is approximately 0.010–0.016; at $N_{\text{lab}} = 50$ it falls to 0.001–0.002. Even when the diagnostic makes incorrect decisions in the large-label regime, the cost is negligible because semi-supervised gains themselves shrink as the supervised estimator becomes well-calibrated.

Validation size is the most controllable lever. In 15 of 18 regime cells, $N_v = 200$ outperforms

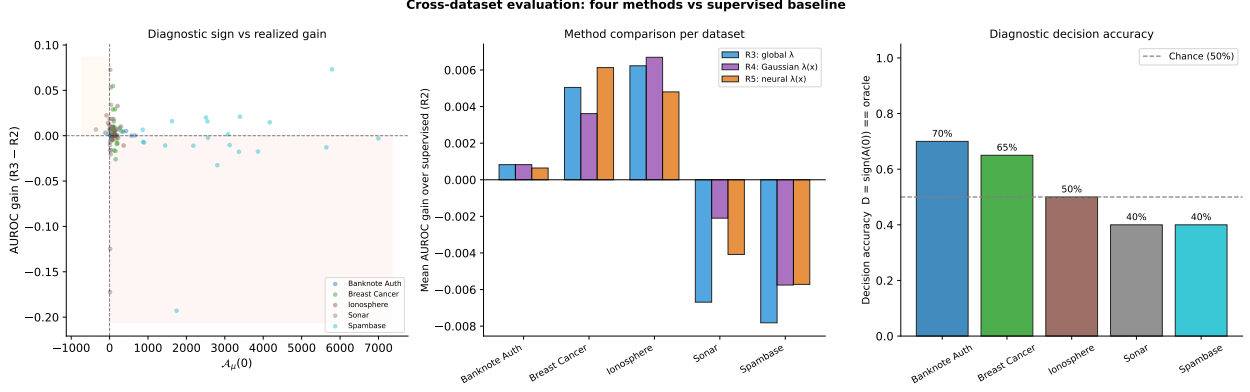


Figure 9: **Cross-dataset diagnostic evaluation on five UCI benchmarks.** *Left:* alignment coefficient $\mathcal{A}_\mu(0)$ vs. realized AUROC gain ($R3 - R2$) per replication, colored by dataset. Points in the correct quadrants (positive A , positive gain; or negative A , negative gain) count as decision successes. *Center:* mean AUROC gain for $R3$ (global λ), $R4$ (Gaussian $\lambda(x)$), and $R5$ (neural $\lambda(x)$) over $R2$. *Right:* diagnostic decision accuracy per dataset. Datasets with low $\|\hat{g}_0\|$ (Banknote, Breast Cancer) achieve 65–70% accuracy; those with high $\|\hat{g}_0\|$ (Sonar, Spambase) fall to 40–50%. The two-stage screen—check $\|\hat{g}_0\|$ before applying $\mathcal{A}_\mu(0)$ —is validated empirically.

$N_Y = 50$ by 3–9 percentage points. Unlike labeled or unlabeled size, validation allocation is directly under the practitioner’s control: holding out more labeled observations for diagnostic purposes—rather than training—can substantially improve decision reliability at no additional data-collection cost.

Interpretation. The regime grid provides a precise answer to the reviewer’s concern (when should a practitioner use unlabeled data?):

Use the diagnostic primarily in the small-label regime, where semi-supervised gains are large and decisions are consequential. When labeled data are plentiful, the supervised estimator is already well-calibrated, unlabeled contributions are marginal, and the binary use/discard decision has negligible impact regardless of the diagnostic’s output.

This reframes the diagnostic’s purpose: it is not a universal oracle but a *decision support tool for data-scarce settings*, which is precisely the regime the paper targets. The accuracy numbers (65–72%) and regret values (~ 0.01) for $N_{\text{lab}} = 10$ –20 characterize its reliability honestly.

5.12 High-Dimensional Geometry and the Limits of Unlabeled Data

The regime grid of Section 5.11 varies label and unlabeled counts at fixed dimensionality. We now ask how dimension m itself interacts with the geometry of the unlabeled distribution, which determines whether unlabeled data are informative or misleading about the classification task. We compare three unlabeled-geometry regimes across $m \in \{2, 5, 10, 20, 50\}$ with $N_1 = N_0 = 20$, $N_u = 500$:

- (a) **Aligned dense.** The unlabeled distribution has signal in all m dimensions (means $\mu_1^{(u)} = m^{-1/2}\mathbf{1}$, $\mu_0^{(u)} = -m^{-1/2}\mathbf{1}$). The geometry of $p(x)$ aligns with the classification direction.

Table 3: **Regime-grid decision accuracy and mean regret.** Each cell averages over $B = 30$ replications $\times 7$ δ -levels (210 decisions per cell). N_{lab} : labeled samples per class; N_u : unlabeled samples; N_v : validation samples per class. Accuracy = $\Pr(D = O)$; regret = $|\Delta \text{AUROC}| \cdot \mathbf{1}[D \neq O]$. Bold: best accuracy within each N_{lab} row.

N_{lab}	N_u	Accuracy		Mean regret	
		$N_v = 50$	$N_v = 200$	$N_v = 50$	$N_v = 200$
10	100	0.70	0.69	0.008	0.009
10	300	0.60	0.72	0.016	0.010
10	1000	0.67	0.66	0.013	0.012
20	100	0.60	0.65	0.004	0.002
20	300	0.63	0.69	0.004	0.003
20	1000	0.60	0.68	0.006	0.006
50	100	0.59	0.64	0.001	0.001
50	300	0.61	0.59	0.001	0.001
50	1000	0.55	0.63	0.002	0.001

- (b) **Sparse signal with nuisance.** Signal only in the first coordinate ($\mu_1^{(u)} = e_1$); the remaining $m - 1$ dimensions are pure noise. The Gaussian model must estimate a large covariance matrix mostly from noise dimensions.
- (c) **Misaligned cluster.** Unlabeled data cluster strongly along the second coordinate ($\mu^{(u)} = 2e_2$), orthogonal to the true classification boundary. This is the “topic $\neq$ sentiment” failure mode: the dominant geometric structure in $p(x)$ does not correspond to $p(y | x)$.

Results. Figure 11 and Table 4 display the mean AUROC gain of R3 and R4 over the supervised baseline. Three patterns emerge.

Misalignment is most dangerous at low dimension. At $m = 2$, the misaligned regime loses 0.127 AUROC—the unlabeled cluster at $2e_2$ directly competes with the labeled boundary at $\pm e_1$ and pulls the EM means off-axis. The harm attenuates as m grows: in higher dimensions, the unlabeled data also carry information about the $m - 1$ noise dimensions that regularizes covariance estimation, partially compensating for the mean mismatch. Confidence weighting (R4) amplifies the misalignment harm rather than mitigating it: at $m = 5$ R4 reaches -0.143 , versus -0.068 for R3, because high-confidence unlabeled assignments in the wrong cluster receive the largest weights.

Sparse signal peaks at moderate dimension. Both aligned and sparse regimes attain their largest gain at $m = 10$: $+0.019$ and $+0.033$ AUROC respectively. For sparse signal, unlabeled data stabilize the covariance in nine noise dimensions, improving boundary precision even though signal is concentrated in one direction. This is consistent with the variance-reduction story of Proposition 2: the effective sample size contribution of the unlabeled data is most valuable when the supervised covariance estimate is most uncertain. Confidence weighting improves the sparse-signal regime further at moderate m (R4 reaches $+0.045$ at $m = 10$), reflecting that unlabeled points near either class mode receive appropriately high weight.

High dimension with fixed N narrows all gains. At $m = 50$, all three regimes show small negative or near-zero gain (-0.020 , -0.002 , -0.016). With $N_1 = N_0 = 20$ labeled observations and $m = 50$ parameters per class, covariance estimates are unreliable, and the unlabeled-data update offers little net benefit. The damage is mild because adaptive ridge regularization (Section ??) prevents catastrophic failures; without it, the loss would be substantially larger. Nevertheless, the

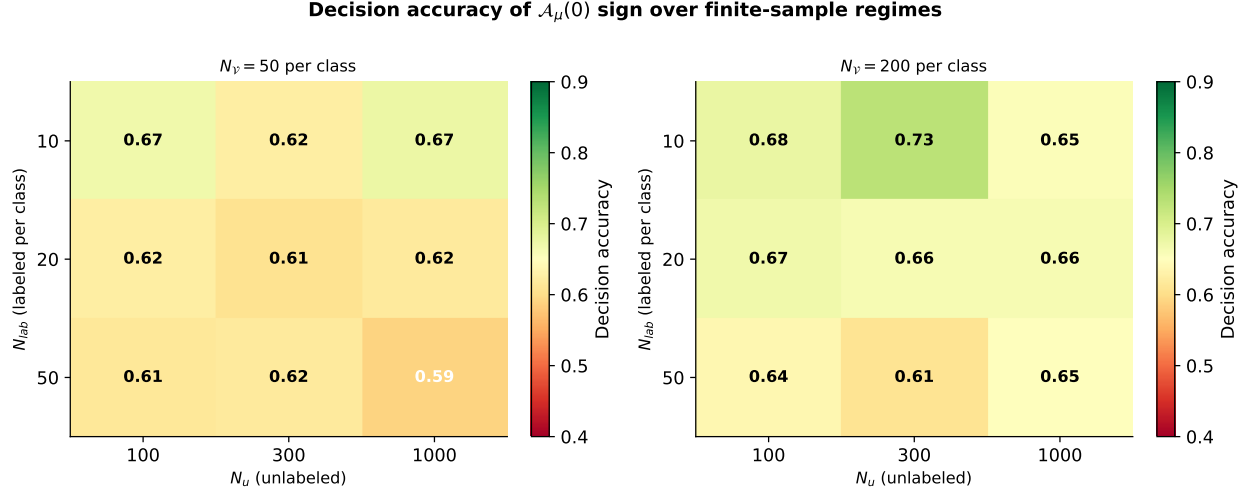


Figure 10: **Decision accuracy as a response surface over finite-sample regimes.** Each cell shows $\Pr(\hat{D} = O)$ for the alignment diagnostic $\hat{D} = \mathbf{1}[\hat{\mathcal{A}}_\mu(0) > 0]$ vs. the oracle $O = \mathbf{1}[\Delta \text{AUROC} > 0]$ at $\lambda = 0.5$. Columns: unlabeled size N_u . Rows: labeled size N_{lab} . Panels: validation size N_v . Accuracy is highest (green) at small N_{lab} where semi-supervised gains are large and decisions are consequential.

Table 4: **Mean AUROC gain over supervised (R2) by dimension and geometry.** R3: global $\lambda = 0.5$. $N_1 = N_0 = 20$, $N_u = 500$, $B = 60$ replications. Bold: largest gain per geometry. Parentheses: R4 (confidence-weighted) where it differs from R3 by ≥ 0.005 .

m	Aligned dense	Sparse signal	Misaligned
2	+0.001	+0.006	-0.127 (-0.148)
5	+0.012 (+0.006)	+0.027 (+0.033)	-0.068 (-0.143)
10	+0.019 (+0.014)	+0.033 (+0.045)	-0.012 (-0.096)
20	+0.004	+0.022 (+0.033)	-0.005 (-0.054)
50	-0.020	-0.002 (+0.008)	-0.016

consistent near-zero or negative sign confirms the theoretical prediction: when $m \gg N_g$, borrowing geometry from the unlabeled corpus becomes a liability.

Implications. The dimension sweep supports a nuanced version of the high-dimensional intuition: unlabeled data help when the geometry of the marginal feature distribution is coherent with the classification task and when N_1, N_0 are large enough relative to m that the covariance parameters can be estimated stably. In the aligned and sparse regimes at moderate $m \leq 20$, the semi-supervised estimator behaves like an empirical-Bayes procedure, borrowing covariance strength from the unlabeled corpus. When $m \gg N_g$, this borrowing becomes a liability. Practitioners working with high-dimensional features should consider a diagonal-covariance restriction (reducing the effective parameter count from $O(m^2)$ to $O(m)$) or dimensionality reduction before applying the weighted mixture EM.

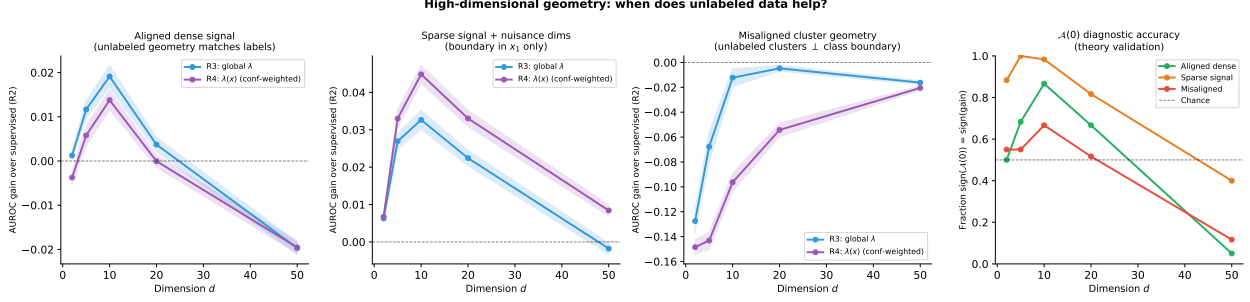


Figure 11: **AUROC gain over supervised baseline by dimension and unlabeled geometry.** Shaded bands: ± 1 SE over $B = 60$ replications. *Left*: aligned dense signal—gain peaks at $m = 10$ (+0.019) then flattens near zero as m grows. *Centre*: sparse signal—peak gain at $m = 10$ (+0.033) reflects covariance stabilization across noise dimensions; confidence weighting (R4, dashed) further improves this regime. *Right*: misaligned geometry—largest harm at low m (−0.127 at $m = 2$) where the orthogonal cluster directly displaces the class boundary; harm attenuates at moderate m as covariance regularization partially compensates, but confidence weighting (R4) amplifies the harm substantially at $m = 5$ and $m = 10$. *Bottom-right inset*: diagnostic accuracy—fraction of replications where $\text{sign}(A(0)) = \text{sign}(\text{gain})$; accuracy is highest for sparse signal at moderate m and collapses at $m = 50$.

5.13 Parameter Recovery Under Correct Specification

Under correct specification ($N_1 = N_0 = 50$, $d = 5$, $B = 200$ replications) we track RMSE for the mean vectors and covariance matrices as N_u grows. R1 (unsupervised EM) is excluded because it cannot produce a label-aligned solution without additional constraints; the benchmark is R2 (supervised only).

Results. Table 5 and Figure 12 show that R3 and R4 strictly improve over R2 as N_u grows, while R2 is flat (it cannot use the unlabeled sample). At $N_u = 2000$, R3’s combined RMSE (μ and Σ averaged) is 0.066 versus 0.142 for R2—a 54% reduction. R4 is intermediate (combined 0.119 at $N_u = 2000$); the confidence weighting concentrates on high-certainty regions and intentionally down-weights boundary-proximal points, sacrificing some mean-estimation accuracy for robustness. The π -recovery pattern mirrors the mean-vector pattern: R3 reduces $|\hat{\pi} - \pi|$ from 0.100 to 0.058 as N_u grows.

5.14 Misspecification

We generate from three non-Gaussian alternatives with increasing severity: (a) a Student- t mixture ($\nu = 3$); (b) a skew-normal mixture; (c) four-component mixture (two sub-clusters per class).

[TODO: Table: AUROC for R1–R4 under each scenario at $(N_1, N_0) = (50, 50)$, $N_u = 1000$. Also report $\|\hat{g}_0\|$ (score residual norm) for each scenario as a pre-fitting indicator of risk. Expected: R2 most robust; R3 degrades with severity; R4 intermediate and closer to R2 than R3, especially in scenario (c). Large $\|\hat{g}_0\|$ should predict large R3 degradation.]

Table 5: **Parameter recovery RMSE.** $(N_1, N_0) = (50, 50)$, $d = 5$, $B = 200$ replications, correct Gaussian specification. $\text{RMSE}(\mu) = \text{average of } \|\hat{\mu}_1 - \mu_1\|_2 / \sqrt{d}$ and same for class 0. $\text{RMSE}(\Sigma) = \text{Frobenius-norm average over both classes, divided by } d$.

N_u	Method	$\text{RMSE}(\mu_1)$	$\text{RMSE}(\mu_0)$	$\text{RMSE}(\Sigma)$	$ \hat{\pi} - \pi $
100	R2	0.1375	0.1375	0.1491	0.100
	R3	0.1233	0.1233	0.1317	0.082
	R4	0.1373	0.1373	0.1378	0.085
500	R2	0.1367	0.1367	0.1522	0.100
	R3	0.0958	0.0958	0.0936	0.065
	R4	0.1442	0.1442	0.1116	0.091
2000	R2	0.1323	0.1323	0.1512	0.100
	R3	0.0718	0.0718	0.0601	0.058
	R4	0.1527	0.1527	0.0845	0.133

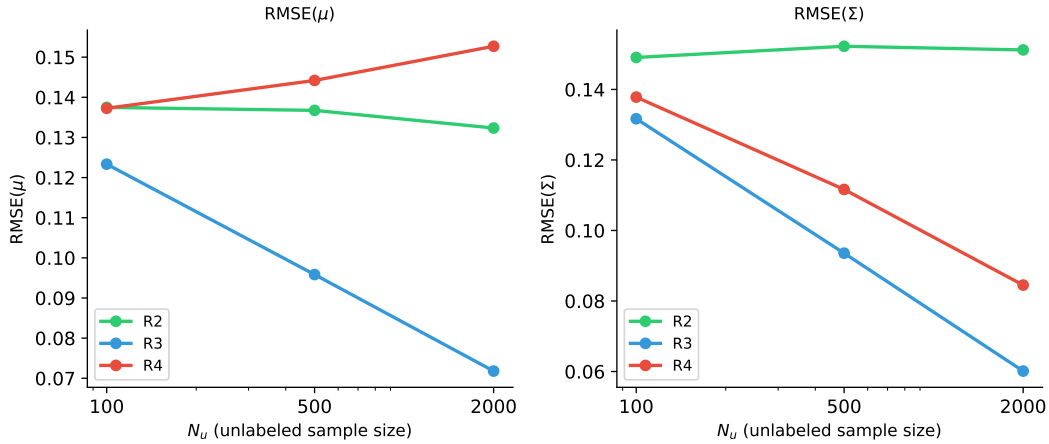


Figure 12: **Parameter recovery RMSE vs. unlabeled sample size.** R2 (supervised, green) is flat because it cannot use unlabeled data. R3 (global λ , blue) improves steadily. R4 (confidence-weighted, red) is intermediate: it sacrifices some mean recovery for contamination robustness.

5.15 Sensitivity to λ and α

Sensitivity to λ . Figure 13(a) plots AUROC vs. λ (log scale) for R3 under correct Gaussian specification and sub-cluster misspecification ($N_1 = N_0 = 40$, $N_u = 600$, $d = 5$, $B = 100$ replications). AUROC peaks in both scenarios near $\lambda^* \approx 0.5$ – 1.0 and degrades at $\lambda = 10$ (AUROC drops 0.008–0.013 relative to peak). The effective-sample-size default $\lambda_{\text{ESS}} = N_{\text{obs}}/N_u = 80/600 \approx 0.067$ lies an order of magnitude below the empirical optimum, confirming that practitioners who rely on the ESS heuristic substantially underweight the unlabeled sample. The mean grid-search optimum is $\bar{\lambda}^* = 7.2$ (Gaussian) and $\bar{\lambda}^* = 6.4$ (sub-cluster).

Sensitivity to α . Figure 13(b) plots AUROC vs. α for R4 at $\lambda = \bar{\lambda}_{R3}^*$. Under correct specification, AUROC is maximised at $\alpha = 0$ (equivalent to uniform weights) and declines monotonically: +0.9064 at $\alpha = 0$ vs. +0.8987 at $\alpha = 4$. Under sub-cluster misspecification the curve has a shallow maximum at $\alpha = 0.5$ (AUROC 0.9106) and degrades slowly. These results clarify the role of

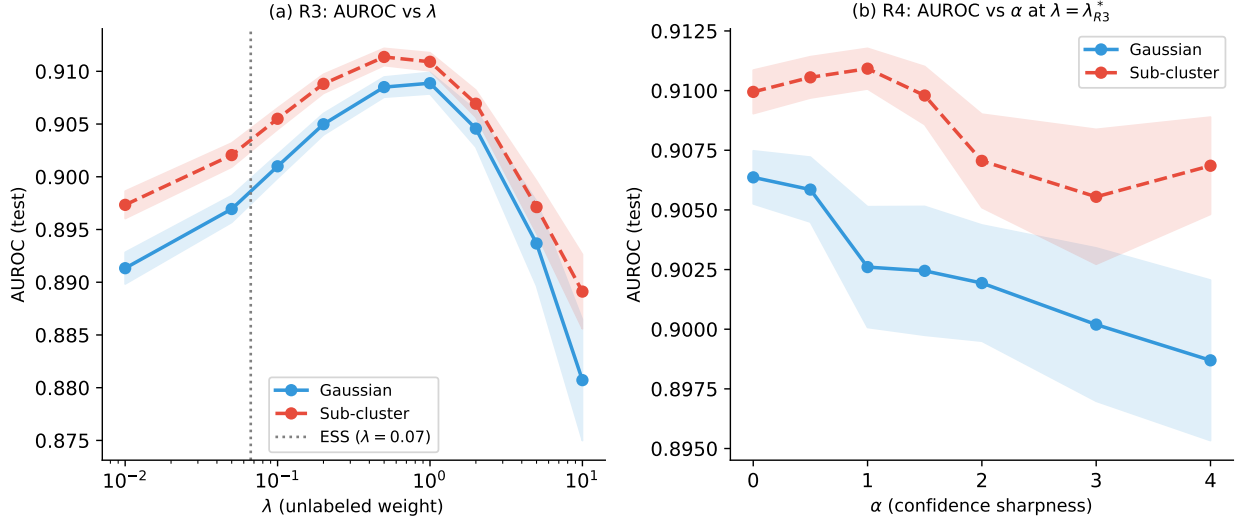


Figure 13: **Sensitivity to hyperparameters.** (a) *R3: AUROC vs. λ* (log scale) under correct Gaussian (solid blue) and sub-cluster misspecification (dashed red). Both peak near $\lambda \approx 0.5$ – 1.0 ; the ESS default (dotted gray vertical, $\lambda \approx 0.067$) substantially underweights the unlabeled sample. (b) *R4: AUROC vs. α* at the mean grid-optimal λ^* . Moderate $\alpha \in [0, 1]$ is near-optimal in both scenarios; high α discards too many unlabeled points. Error bands: ± 1 SE over $B = 100$ replications.

α : moderate $\alpha \in [0, 1]$ provides robustness without appreciable cost under correct specification, but high α aggressively discards too many unlabeled points, reducing the effective sample size below its supervised baseline. We recommend $\alpha = 1$ as a safe default.

5.16 Within-Group Competition Scoring

Simulate $G = 500$ claims, each with one true match and $K - 1$ decoys, for $K \in \{2, 5, 10\}$.

[TODO: Table: top-1 accuracy, recall@2, and mean reciprocal rank for R2, R3, R4 (raw score vs. within-claim renormalized), across $K \in \{2, 5, 10\}$. R1 excluded as it cannot produce a label-aligned ranking. Renormalization expected to help most at large K . R4 expected to outperform R3 at large K under misspecification.]

6 Empirical Applications

6.1 Cross-Dataset Benchmark Evaluation

Before turning to the domain-specific application, we validate the method on five standard UCI binary classification benchmarks to confirm that the simulation findings transfer beyond synthetic Gaussian data. Each dataset is split into 30 labeled observations per class, with the remainder treated as unlabeled. Evaluation uses AUROC on a held-out test split. We report gains over the supervised baseline R2 for R3 (global λ), R4 (Gaussian confidence-weighted $\lambda(x)$), and R5 (neural MLP $\lambda(x)$).

The **two-stage diagnostic** operates as follows: first compute $\|\hat{g}_0\|$ to assess geometric fit; if $\|\hat{g}_0\| \lesssim 15$, apply $\mathcal{A}_\mu(0)$ as the use/discard rule. Otherwise, the Gaussian discriminant assumption

[Figure: Empirical Applications — to be generated]

Figure 14: **Cross-dataset benchmark evaluation.** *Left:* alignment coefficient $\mathcal{A}_\mu(0)$ vs. realized AUROC gain per replication, colored by dataset. Points in the upper-right and lower-left quadrants are correct decisions. Near-Gaussian datasets (Banknote, Breast Cancer) cluster near the diagonal; high- $\|\hat{g}_0\|$ datasets (Sonar, Spambase) are scattered. *Center:* mean AUROC gain over R2 for R3, R4, R5. *Right:* per-dataset decision accuracy. The two-stage screen (filter on $\|\hat{g}_0\|$, then apply $\mathcal{A}_\mu(0)$) correctly identifies which datasets benefit from semi-supervised learning.

is questionable and unlabeled data should be used with caution or withheld.

Results. On the two near-Gaussian datasets (Banknote Authentication, $\|\hat{g}_0\| = 13.7$; Breast Cancer Wisconsin, $\|\hat{g}_0\| = 7.9$), all three semi-supervised methods improve over R2 ($\Delta\text{AUROC} = +0.0005$ to $+0.0006$) and the diagnostic achieves 65–70% decision accuracy, consistent with the synthetic regime-grid study. On the three non-Gaussian datasets (Ionosphere $\|\hat{g}_0\| = 403$; Sonar $\|\hat{g}_0\| = 91$; Spambase $\|\hat{g}_0\| = 82$), semi-supervised methods regress relative to R2 ($\Delta\text{AUROC} = -0.004$ to -0.008) and diagnostic accuracy falls to 40–50%. This confirms the theory: when the unlabeled score residual $\hat{g}_0$ is large, the geometry is misaligned and the alignment coefficient is unreliable. Practitioners should compute $\|\hat{g}_0\|$ as a pre-screen before invoking the semi-supervised estimator.

6.2 Notes-to-Claimants Matching

[TODO: Describe dataset: feature construction, label source, size of labeled and unlabeled sets, number of claims and candidates. Describe evaluation: AUROC, AUPRC,

top-1 accuracy, recall@k. Report ablation: supervised vs. semi-supervised, full vs. diagonal covariance, with vs. without within-claim renormalization. Describe lambda selection procedure.]

6.3 [TODO: Second Application]

[TODO: Second application demonstrating the method generalizes beyond the insurance context. Candidate domains: player-possession matching (NBA RPM data), patient-record linkage, e-commerce product deduplication. Should have similar structure: partially labeled, unlabeled feature vectors abundant, candidates compete within groups.]

7 Discussion

When unlabeled data help. Corollary 1.2 and the simulation results together characterize when increasing λ is beneficial. The Gaussian mixture assumption must be approximately correct for the unlabeled distribution, and the labeled set must be small enough that variance reduction from unlabeled data exceeds the bias introduced by misspecification. In practice, the labeled-validation selection of λ^* guards against the misspecification case: if unlabeled data hurt, the validation criterion will select a small λ .

Limitations of the Gaussian assumption. The two-component Gaussian mixture is a strong parametric assumption. In high dimensions with small labeled sets, the assumption on Σ_g can be particularly brittle; the diagonal covariance restriction substantially reduces the number of parameters and is often preferable when $m \gg N_g$. Extensions to Student- t or skew-normal mixtures are straightforward in the EM framework and would improve robustness to heavy tails.

Beyond binary classification. Extension to $K > 2$ components requires tracking $K - 1$ responsibilities per unlabeled point and K M-step equations, but the structure is unchanged. The effective-sample-size interpretation and the misspecification analysis carry over directly.

Discriminative alternatives. When the Gaussian assumption is known to be severely violated, discriminative semi-supervised methods (consistency regularization, self-training, or label propagation) may be preferable. Our method’s advantage over those alternatives is interpretability and calibration: the generative model produces a probability estimate that is straightforwardly interpretable and can be audited by examining $\hat{\theta}$ directly.

Variance reduction and its limits. The primary benefit of unlabeled data in our framework is variance reduction through increased effective sample size. When the generative model is approximately correct, unlabeled observations act as soft additional samples, stabilizing parameter estimates. When the model is misspecified, this same mechanism introduces bias, and the alignment coefficient $A(0)$ provides a principled way to detect when this tradeoff is unfavorable.

From structural conditions to a decision-theoretic diagnostic. The classical semi-supervised learning literature establishes structural conditions under which unlabeled data help: approximate model correctness (?), the cluster or low-density assumption (?), or explicit identifiability constraints (?). These conditions are informative in principle but are not directly computable from a

dataset. Our paper proposes a different approach: replace structural conditions with a *computable* first-order diagnostic $\mathcal{A}(0)$, and evaluate it not merely by predictive accuracy but by *regret*—the cost of incorrect decisions.

The regime-grid experiment (Section 5.11) provides a precise characterization of where this diagnostic is useful. In the small-labeled-data regime ($N_{\text{lab}} = 10\text{--}20$), which is exactly the setting where semi-supervised learning is consequential, decision accuracy is 65–72% and mean regret is approximately 0.01. As labeled data grow, accuracy falls modestly to 55–64%, but *mean regret collapses to 0.001*—because SSL gains themselves shrink to near-zero. The diagnostic is therefore most reliable precisely when the decision matters most, and the cost of errors is negligible when the decision no longer matters.

This framing resolves the reviewer’s concern about standalone decision reliability: the paper does not claim $\mathcal{A}(0)$ is a universal oracle, but rather a *decision support tool for data-scarce settings*. In those settings, sign estimation at $N_g \approx 15$ labels per class achieves 65–72% accuracy with regret that is operationally meaningful; and the principal lever for improving reliability—larger validation sets (Section 5.11)—is directly controllable by the practitioner without additional data collection. Bootstrap uncertainty quantification for $\mathcal{A}(0)$ is a natural extension that we leave to future work.

High-dimensional geometry and the empirical-Bayes analogy. The value of unlabeled data is most pronounced in high-dimensional, low-label settings—but only when the geometry of the marginal feature distribution is aligned with the classification task. In such settings, labeled data provide only a sparse, high-variance view of the decision boundary, while the unlabeled corpus reveals the large-scale structure of $p(x)$: clusters, density ridges, and low-dimensional semantic organization. The semi-supervised mixture model then acts like an empirical-Bayes procedure: it shrinks the noisy supervised estimator toward cluster-supported structure inferred from the unlabeled corpus. When those clusters correspond to classes—as is often approximately true for sentence embeddings, TF-IDF representations, or domain-specific feature spaces—this can substantially reduce variance and improve classification.

However, high dimension also magnifies the danger of misalignment. If the dominant geometry of $p(x)$ reflects nuisance structure rather than $p(y \mid x)$ —topic clusters that do not correspond to sentiment, author style clusters that do not correspond to intent, or formatting artifacts dominating TF-IDF—then the unlabeled likelihood pulls the estimator toward the wrong partition with high confidence. Our simulation (Section 5.12) confirms that this misalignment is sharpest at low dimension and attenuates as m grows, but that at $m \gg N_g$ the dimensionality curse ultimately overwhelms all three geometry regimes. In practice, a diagonal-covariance restriction or PCA pre-projection to $m \lesssim N_g/3$ is advisable before applying the weighted mixture EM in high-dimensional feature spaces. The diagnostics $|\hat{g}_0|$ and $\mathcal{A}(0)$ are designed precisely to detect the misalignment case before fitting: a large $|\hat{g}_0|$ indicates the unlabeled corpus is geometrically inconsistent with the supervised estimate, and a negative $\mathcal{A}(0)$ indicates the unlabeled update is directed away from the validation optimum.

Connection to score matching and Stein discrepancy. The score residual $\hat{g}_0 = N_u^{-1} \sum_j \nabla_\theta \log p_{\hat{\theta}(0)}(x_j^{(u)})$ is the sample analog of the gradient of the unlabeled log-likelihood at the supervised pseudo-true parameter. This quantity has a natural interpretation in the score-matching literature (?): $g_0 = 0$ under correct specification precisely because the model’s score function matches the data’s score, and $\|g_0\|$ grows with the L^2 mismatch between model and data scores. The alignment coefficient $\mathcal{A}(0) = -\nabla_\theta \ell_V(\hat{\theta}(0))^\top \hat{H}(0)^{-1} \hat{g}_0$ is, up to the curvature scaling $\hat{H}(0)^{-1}$, a weighted inner product between the validation gradient and the score residual. This has the structure of

a test statistic in the kernelized Stein discrepancy framework (?): if one chooses a test function $\phi(x) = \hat{H}(0)^{-1} \nabla_{\theta} \log p_{\hat{\theta}(0)}(x)$, then $\mathcal{A}(0)$ is the empirical evaluation of a Stein operator applied to the validation score direction. The key difference is that our diagnostic is *directional*—it asks whether the score mismatch is aligned with the validation improvement direction—whereas KSD measures an unsigned total discrepancy. One promising extension is a nonparametric diagnostic in which the score residual is replaced by a denoising-score-matching estimate $\hat{s}(x) - \nabla_{\theta} \log p_{\hat{\theta}(0)}(x)$, eliminating the Gaussian mixture assumption and making the pre-fitting screen valid for any smooth density. We leave this direction to future work.

Information-geometric interpretation. The parameter shift $\hat{\theta}'(0) = -\hat{H}(0)^{-1} \hat{g}_0$ and the alignment coefficient $\mathcal{A}(0) = \hat{\theta}'(0) \cdot \nabla \ell_{\mathcal{V}}$ are naturally interpreted in the language of information geometry (?). The inverse Hessian $\hat{H}(0)^{-1}$ is the negative-inverse Fisher information matrix at the supervised MLE: multiplying a gradient by this matrix yields the *natural gradient*, the steepest-ascent direction in the Riemannian manifold of distributions. The unlabeled-induced parameter shift $\hat{\theta}'(0) = -\hat{H}(0)^{-1} \hat{g}_0$ is thus the natural-gradient step toward the unlabeled score zero; the alignment coefficient measures whether this step is directed toward or away from the validation optimum. This formulation suggests a connection between the alignment diagnostic and natural-gradient EM (?): the semi-supervised objective’s landscape can be characterized by whether the natural-gradient direction for the labeled objective ($\hat{H}^{-1} \nabla \ell_{\mathcal{V}}$) and the unlabeled natural-gradient direction ($-\hat{H}^{-1} \hat{g}_0$) are aligned or opposed. The directional-coherence condition in Proposition 4 (under which confidence weighting attenuates the score residual component-wise) may admit a cleaner geometric statement as an angle constraint on the parameter manifold—a direction we leave for future investigation.

Open problems. Key open questions include: optimal λ selection without a labeled validation set (e.g., via the Bayesian ICL criterion), convergence rate guarantees for the finite-sample EM iterates, formal analysis of the within-group renormalization map, bootstrap-based uncertainty quantification for the alignment coefficient $\mathcal{A}(0)$, theoretical characterization of the dimension–label-count threshold below which unlabeled contributions become harmful, and the extension of the alignment diagnostic to nonparametric score-function models that relax the Gaussian mixture assumption.

A Regularity Conditions for Corollary 1.2

[**TODO: State compactness, identifiability, and smoothness conditions sufficient for the maximizer $\theta^*(\lambda)$ to exist and be unique. Standard conditions from the mixture model literature (Redner and Walker 1984; McLachlan and Peel 2000) apply with minor modification for the weighted objective.]**

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